

Electronics and Communication Systems

Electronics Systems

Master Degree in **Computer Engineering**

Luca Fanucci

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Email: Pietro.nannipieri@unipi.it

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VHDL Hands-on Lab Exercises



With the support of:

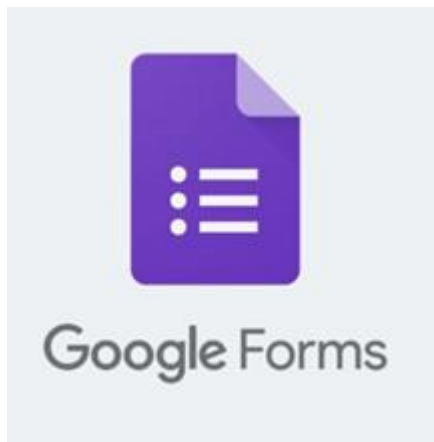
Tommaso Bocchi

Phone: +39 050 2217 625

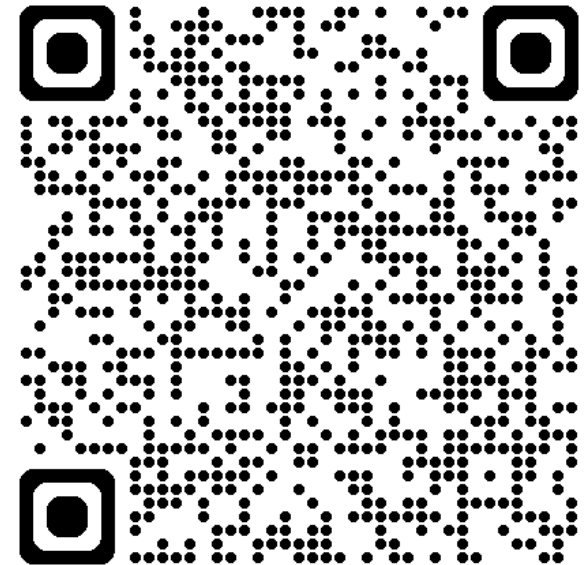
Email: tommaso.bocchi@phd.unipi.it

Students' background and Contacts

Name Surname	Telephone	E-mail	Home Town/Country	Bachelor Degree/ University
Luca Fanucci	347 5013837	luca.fanucci@unipi.it	Montecatini Terme (PT) Italy	Electronic Engineering University of Pisa



<https://forms.gle/caTJApMHYxh2CiHg6>



Course Schedule

Computer Eng. M1 21 – 25 set 2026					
Tutto il giorno	lun 21/9	mar 22/9	mer 23/9	gio 24/9	ven 25/9
08:30	08:30 - 11:30 PERFORMANCE EVALUATION OF COMPUTER SYSTEMS AND NETWORKS COMPUTER ENGINEERING (1) Etr F2 G. NARDINI, G. STEA				08:30 - 10:30 PERFORMANCE EVALUATION OF COMPUTER SYSTEMS AND NETWORKS COMPUTER ENGINEERING (1) Ing S15
09:30					
10:30				10:30 - 12:30 QUANTUM COMPUTING AND QUANTUM INTERNET COMPUTER ENGINEERING (1) Ing S13 L. LENZINI	
11:30	11:30 - 13:30 ELECTRONICS AND COMMUNICATIONS SYSTEMS - COMMUNICATIONS SYSTEMS COMPUTER ENGINEERING (1) Etr F2	11:30 - 13:30 LARGE-SCALE AND MULTI-STRUCTURED DATABASES COMPUTER ENGINEERING (1) Ing S15 P. DUCANGE	11:30 - 13:30 LARGE-SCALE AND MULTI-STRUCTURED DATABASES COMPUTER ENGINEERING (1) Ing C31 P. DUCANGE	11:30 - 13:30 QUANTUM COMPUTING AND QUANTUM INTERNET COMPUTER ENGINEERING (1) Ing ADI11	11:30 - 13:30 LARGE-SCALE AND MULTI-STRUCTURED DATABASES COMPUTER ENGINEERING (1) Ing A28 P. DUCANGE
12:30					
13:30				13:30 - 16:30 ELECTRONICS AND COMMUNICATIONS SYSTEMS - COMMUNICATIONS SYSTEMS COMPUTER ENGINEERING (1) Etr F4 L. FANUCCI, M. MORETTI	
14:30	14:30 - 17:30 ELECTRONICS AND COMMUNICATIONS SYSTEMS - COMMUNICATIONS SYSTEMS COMPUTER ENGINEERING (1) Etr F2 L. FANUCCI, M. MORETTI	14:30 - 16:30 LARGE-SCALE AND MULTI-STRUCTURED DATABASES COMPUTER ENGINEERING (1) Ing S15 P. DUCANGE	15:30 - 18:30 PERFORMANCE EVALUATION OF COMPUTER SYSTEMS AND NETWORKS COMPUTER ENGINEERING (1) Ing S11 G. NARDINI, G. STEA	16:30 - 18:30 QUANTUM COMPUTING AND QUANTUM INTERNET COMPUTER ENGINEERING (1) Ing S13 L. LENZINI	15:30 - 18:30 ELECTRONICS AND COMMUNICATIONS SYSTEMS - COMMUNICATIONS SYSTEMS COMPUTER ENGINEERING (1) Ing S15 L. FANUCCI, M. MORETTI, P. NANNIPERI
15:30					
16:30					
17:30					

Question Time

by appointment:

Tel. 347 5013837

Email: luca.fanucci@unipi.it



Tel. 050 2217625

Email: pietro.nannipieri@unipi.it



Tel. 050 2217625

Email: tommaso.bocchi@phd.unipi.it



Course Goals and Prerequisites

▣ Goals

- ▣ Provide you with an introduction to the **techniques** and **methodologies** used to produce embedded systems based on Very Large Scale Integration (VLSI) technology
- ▣ **Circuits** and **Architectures** for **basic digital signal processing blocks** to be used for complex system synthesis according to given **performance criteria**: area, speed, power consumption and reliability.
- ▣ Give you some **hands-on experience** with software tools used for Computer Aided Design (CAD)
- ▣ Create a foundation for further exploration of VLSI

▣ Prerequisites

- ▣ Basic understanding of circuits and electronics
- ▣ Basic understanding of logic design

Course Goals and Prerequisites

- Prerequisites
 - Basic understanding of circuits and electronics
 - Basic understanding of logic design

Prerequisiti-Reti Logiche - Prof. Stea (1/2)

LINGUAGGIO ASSEMBLER: Descrizione Linguaggio macchina, mnemonico, assembler. Indirizzamento degli operandi. istruzioni operative: trasferimento, aritmetiche, traslazione/rotazione, logiche. Istruzioni di controllo. Istruzioni per operandi stringa. Assemblaggio e debugging. Utilizzo del debugger GDB. Sottoprogrammi e passaggio dei parametri.

RETI COMBINATORIE: Le porte AND, OR, NOT, NAND e NOR; il decodificatore/demultiplicatore; il moltiplicatore. Le porte a tre strati e le loro applicazioni. Modalità di descrizione, trattazione algebrica e sintesi ottima delle reti combinatorie.

RETI SEQUENZIALI ASINCRONE: I flip-flop SR, D latch e D edge-triggered. Le memorie RAM.

Prerequisiti-Reti Logiche - Prof. Stea(2/2)

RETI SEQUENZIALI SINCRONIZZATE: L'elemento di registro; i registri in traslazione e i contatori. Reti sequenziali sincronizzate di Moore, di Mealy e di Mealy Ritardato: modelli funzionali, modalità di descrizione, modelli implementativi. Il flip-flop J-K. Reti sequenziali complesse: descrizione in un linguaggio di trasferimento tra registri, sintesi in accordo al modello strutturale con parte operativa e parte controllo (con particolare riferimento ai modelli microprogrammati).

STRUTTURA FISICA DI UN CALCOLATORE: Moduli di base e loro collegamento. Struttura interna del processore; della memoria e di alcune interfacce (parallele, seriali, di conteggio e per la conversione A/D e D/A). L'ingresso/uscita dati a controllo di programma. Cenni al meccanismo di interruzione.

ALGORITMI E RETI DI TIPO ARITMETICO: Richiami sulla rappresentazione dei numeri naturali, interi e reali; gli algoritmi e le reti fondamentali per una aritmetica dei numeri naturali e dei numeri interi.

Prerequisiti-Elettronica Digitale-Prof. Piotto

PORTE LOGICHE IN TECNOLOGIA CMOS: Inverter CMOS, parametri caratteristici dei circuiti digitali, calcolo dell'energia dissipata durante la commutazione, fan-in, fan-out, sintesi delle porte logiche tramite il metodo della pull-up/pull-down network, protezione ingressi da scariche elettrostatiche, logica a pass transistor, body effect.

CIRCUITI LOGICI SEQUENZIALI: Latch realizzato con inverter, flip-flop S/R in tecnologia CMOS, flip-flop D e D edge triggered, il circuito integrato NE555 e le sue applicazioni.

MEMORIE A SEMICONDUCTORE: Struttura a matrice delle memorie, ROM a diodi, ROM e PROM a MOS, memoria SRAM e DRAM, principio di funzionamento del decoder degli indirizzi (wired NOR), memorie EPROM e EEPROM.

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[WCN-LM] COMPUTER ENGINEERING

LAUREA MAGISTRALE INGLESE

Anno di immatricolazione 2026/2027[Scheda del corso](#)

Scheda del corso

Informazioni generali

Durata: 2 anni**Ordinamento:** 2026**Data di approvazione:** 30/06/2025**stato:** Chiuso**Crediti:**

- comune: undefined CFU
- COMPUTER SYSTEMS AND NETWORKS : 120 CFU
- CYBER-PHYSICAL SYSTEMS: 120 CFU
- CYBERSECURITY: 120 CFU

Lingua: Inglese**Classe di laurea:** LM-32 - Classe delle lauree magistrali in Ingegneria informatica**Accesso:** Libero**Dipartimento:** DIPARTIMENTO DI INGEGNERIA DELL'INFORMAZIONE**Sito web:** <https://computer.ing.unipi.it/ce-lm>

Course Main Topics

- ▣ Design metrics in digital IC design for telecommunication systems
- ▣ Design space exploration trading-offs among General Purpose Computer, Digital Signal Processor, Application Specific Instruction Set Processor, Field Programmable Gate e Application Specific Integrated Circuit.
- ▣ Profiling and mapping of DSP algorithms into VLSI architectures including finite arithmetic effects (floating vs. fixed point implementation).
- ▣ CMOS implementation of main digital signal processing blocks: adder, multiplier, arithmetic logic unit, multiplier & accumulator unit, CORDIC, FIR and IIR filters.
- ▣ High-level electronic design automation tools for hardware-software co-design for semi-custom digital IC. Hardware description languages modeling, simulation and logic synthesis.
- ▣ Power Consumption in CMOS digital IC. Design techniques for the reduction of power consumption at different level of abstractions.

Course Material (1/5)

< Tutti i team



Electronics Systems

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598II 26/27 - ELECTRONICS AND

Pagina iniziale

Class Notebook

Il lavoro in classe

Attività

Voti

Insights

Reflect

▼ Canali principali

Generale

Communications Systems

Electronics Systems



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↳ Lecture Slides

2 minutes ago

Luca Fanucci



↳ VHDL and Hands-on-Exercises

About a minute ago

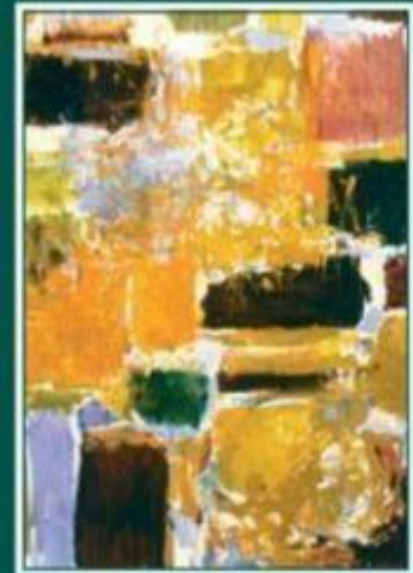
Luca Fanucci

Course Material (2/5)

- CMOS Design Bibliographic References:
 - Weste and Eshraghian, *"Principles of VLSI Design – A Systems Perspective (2nd ed)"*, Addison Wesley
 - M. Smith, *"Application Specific Integrated Circuits"*, Addison-Wesley, 1997
 - J. M. Rabaey, *"Digital Integrated Circuits - A Design Perspective"*, Prentice Hall

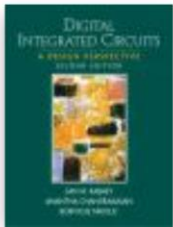
<http://icbook.eecs.berkeley.edu/overview>

DIGITAL INTEGRATED CIRCUITS A DESIGN PERSPECTIVE SECOND EDITION



JAN M. RABAEY
ANANTHA CHANDRAKASAN
BORIVOJE NIKOLIC

Course Material (3/5)



LIBRO

Digital integrated circuits : a design perspective / Jan M. Rabaey, Anantha Chandrakasan, Borivoje Nikolic

Nikolic, Borivoje ; Chandrakasan, Anantha P ; Rabaey, Jan M

Upper Saddle River, NJ Pearson education international : Prentice Hall ; 2003

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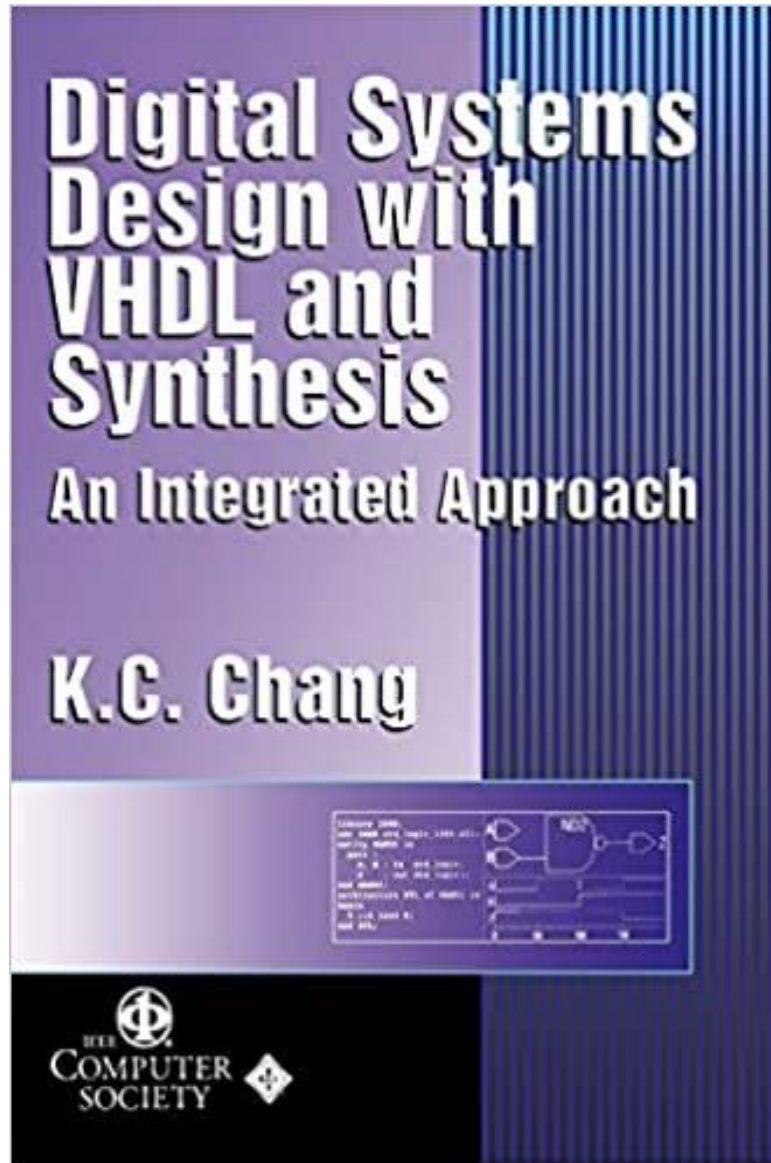
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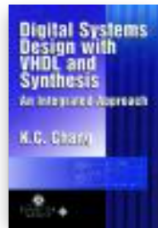
(1 copia, 1 disponibile, 0 richieste)

Course Material (4/5)

- VHDL Bibliographic References:
 - IEEE Standard VHDL Language Reference Manual : IEEE Std 1076-1993
 - S. Mazor, P. Langstraat, *"A guide to VHDL"*, Kluwer Academic Publishers
 - Douglas L. Perry, *"VHDL" (2nd edition)*, McGraw Hill
 - B. Cohen, *"VHDL Coding Styles and Methodologies"*, Kluwer Academic Publishers
 - K.C. Chang, *"Digital Systems Design with VHDL and Synthesis"*, IEEE Computer Society
 - W.F. Lee, *"VHDL: Coding and Logic Synthesis with SynopsysTM"*, Academic Press



Course Material (5/5)



LIBRO

Digital systems design with VHDL and synthesis : an integrated approach / K.C. Chang
Chang, Kou Chuan 1957-

Los Alamitos Ca : IEEE computer society ; 1999

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VHDL Simulation & Synthesis (1/3)

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Questa*-Intel® FPGA Edition Software

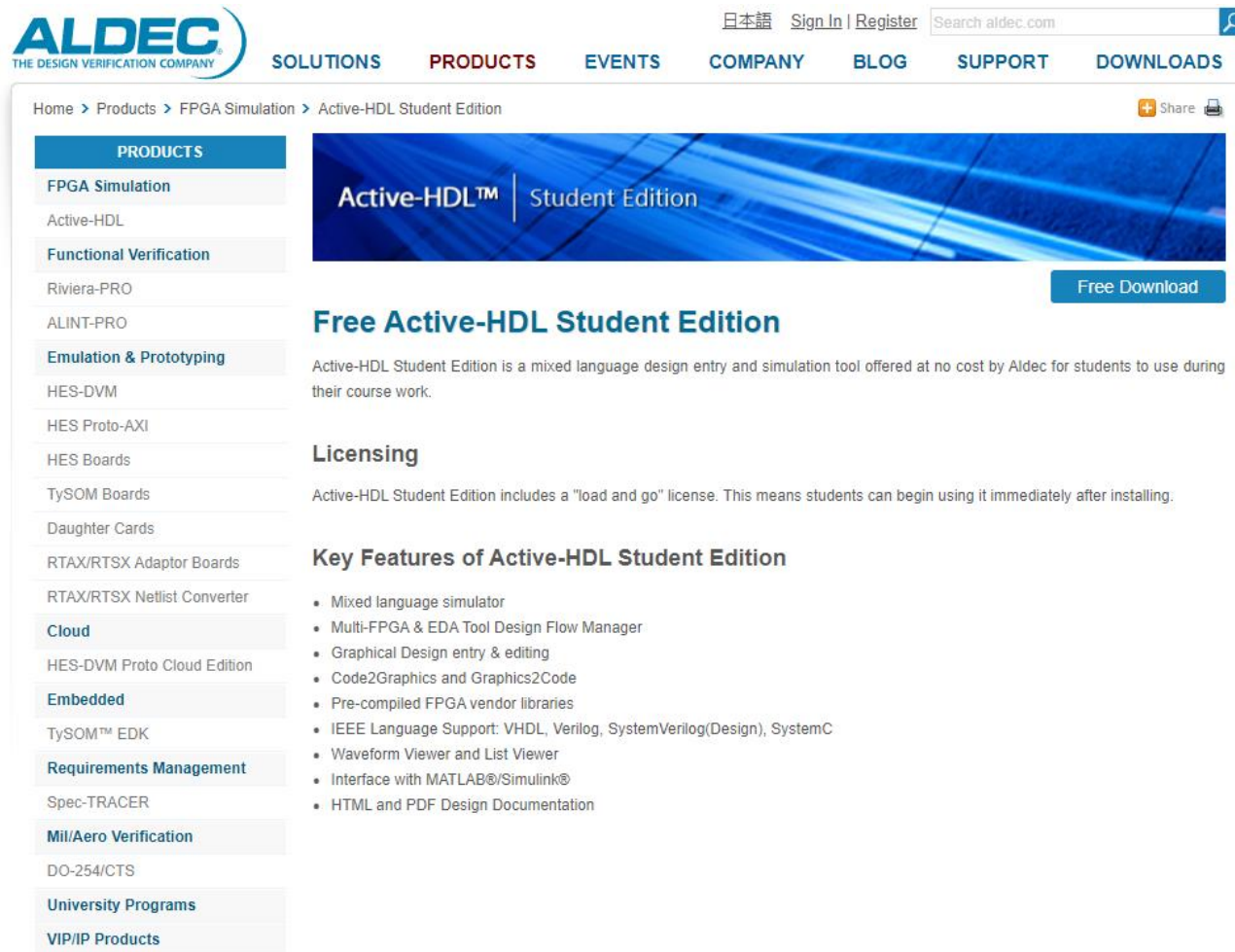
[Overview](#)[FAQ](#)[Support](#)[Feedback](#)

Both Questa*-Intel® FPGA and Questa*-Intel® FPGA Starter software editions are a version of the Siemens EDA Questa* Core software targeted for Intel® FPGAs devices. Both editions support the below:

- Intel® Quartus® Prime Lite Edition, Standard edition, and Pro edition, Intel® High Level Synthesis Compiler, Intel oneAPI and DSP Builder for Intel® FPGAs software.
- All Intel® FPGAs and Intel® MAX® CPLDs.
- Intel gate-level libraries (includes behavioral simulation, HDL test benches, and Tcl scripting).
- 64-bit [Operating System \(OS\) Windows and Linux](#).
- Questa*-Intel® FPGA Edition software supports designs of up to 5,000 instances. Note that instances from Intel pre-compiled libraries do not count towards the 5,000 instance limit.

<https://www.intel.com/content/www/us/en/software/programmable/quartus-prime/questa-edition.html>

VHDL Simulation & Synthesis (2/3)



The screenshot shows the Aldec website's product page for Active-HDL Student Edition. The header includes the Aldec logo, navigation links (SOLUTIONS, PRODUCTS, EVENTS, COMPANY, BLOG, SUPPORT, DOWNLOADS), and a search bar. The breadcrumb trail is: Home > Products > FPGA Simulation > Active-HDL Student Edition. A sidebar on the left lists various products under categories like PRODUCTS, Emulation & Prototyping, Cloud, Embedded, Requirements Management, and Mil/Aero Verification. The main content area features a blue banner with the product name, a 'Free Download' button, and sections for 'Free Active-HDL Student Edition', 'Licensing', and 'Key Features of Active-HDL Student Edition'.

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SOLUTIONS PRODUCTS EVENTS COMPANY BLOG SUPPORT DOWNLOADS

Home > Products > FPGA Simulation > Active-HDL Student Edition

PRODUCTS

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- Functional Verification
 - Riviera-PRO
 - ALINT-PRO
- Emulation & Prototyping
 - HES-DVM
 - HES Proto-AXI
 - HES Boards
 - TySOM Boards
 - Daughter Cards
 - RTAX/RTSX Adaptor Boards
 - RTAX/RTSX Netlist Converter
- Cloud
 - HES-DVM Proto Cloud Edition
- Embedded
 - TySOM™ EDK
- Requirements Management
 - Spec-TRACER
- Mil/Aero Verification
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- University Programs
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Active-HDL™ | Student Edition

Free Download

Free Active-HDL Student Edition

Active-HDL Student Edition is a mixed language design entry and simulation tool offered at no cost by Aldec for students to use during their course work.

Licensing

Active-HDL Student Edition includes a "load and go" license. This means students can begin using it immediately after installing.

Key Features of Active-HDL Student Edition

- Mixed language simulator
- Multi-FPGA & EDA Tool Design Flow Manager
- Graphical Design entry & editing
- Code2Graphics and Graphics2Code
- Pre-compiled FPGA vendor libraries
- IEEE Language Support: VHDL, Verilog, SystemVerilog(Design), SystemC
- Waveform Viewer and List Viewer
- Interface with MATLAB®/Simulink®
- HTML and PDF Design Documentation

https://www.aldec.com/en/products/fpga_simulation/active_hdl_student

VHDL Simulation & Synthesis (3/3)

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AMD Vivado™ Design Suite

The Vivado 2026.1 release is now available, featuring a new tiered licensing model for greater flexibility, new device support, major NoC and QoR improvements, and ease-of-use enhancements for designing with AMD Versal™ adaptive SoCs.

[Download Now](#)[Versal Features](#)[Overview](#)[What's New](#)[Advantages](#)[Design Flows](#)[Resources](#)

AMD Vivado™ Design Suite Moving to More Flexible Licensing Options

Starting with the 2026.1 release, AMD Vivado™ Design Suite is evolving to a new tiered licensing model with more licensing options at more price points, providing flexibility to pay only for the device families and features that you need.

Choose the license tier that fits your design needs, from free entry-level access to advanced, full-featured software environments, while Vivado Enterprise continues unchanged.

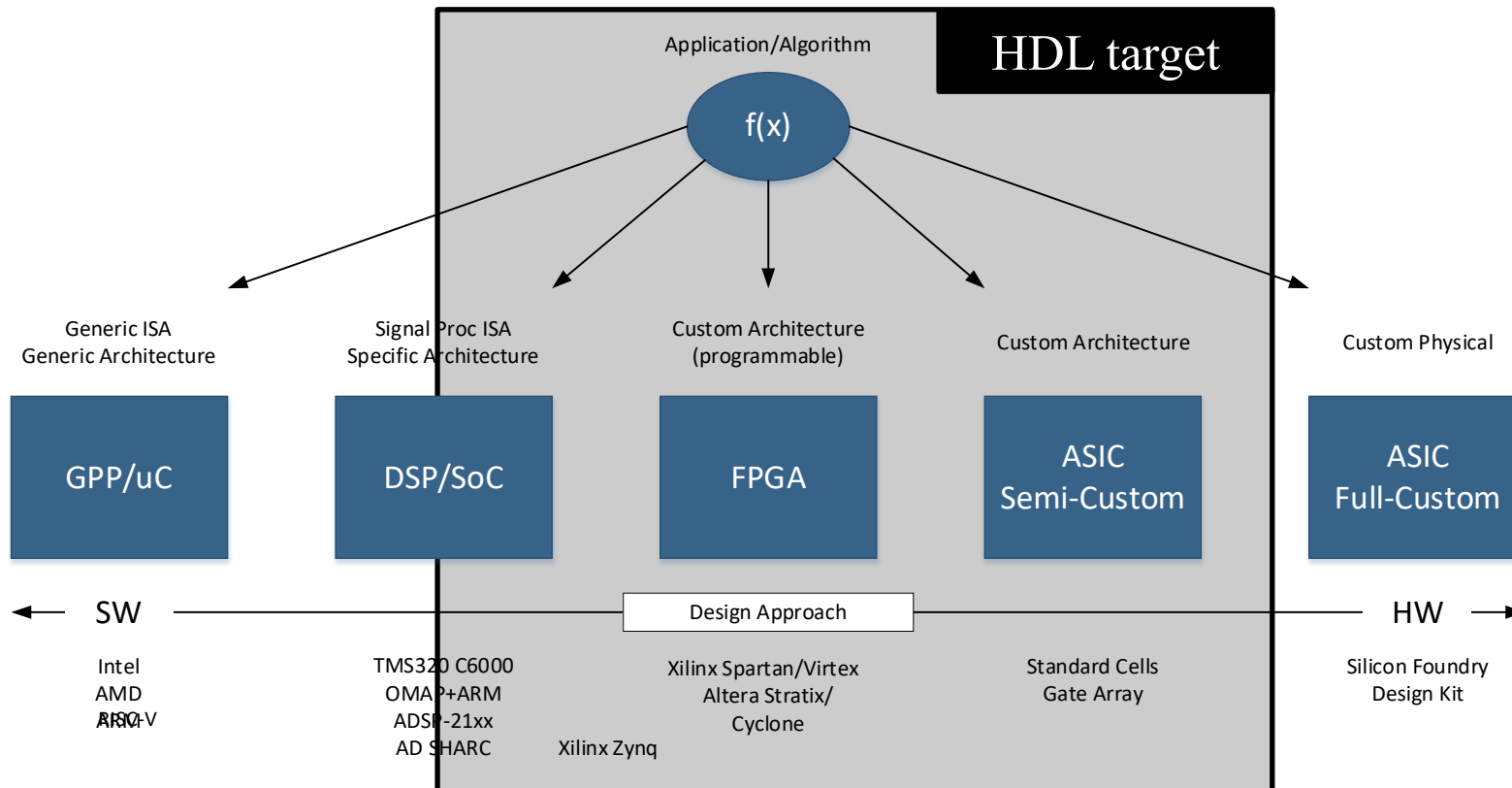
[Learn More >](#)

VHDL Logic synthesis and FPGA Programming

<http://www.xilinx.com/products/design-tools/vivado/vivado-webpack.html>

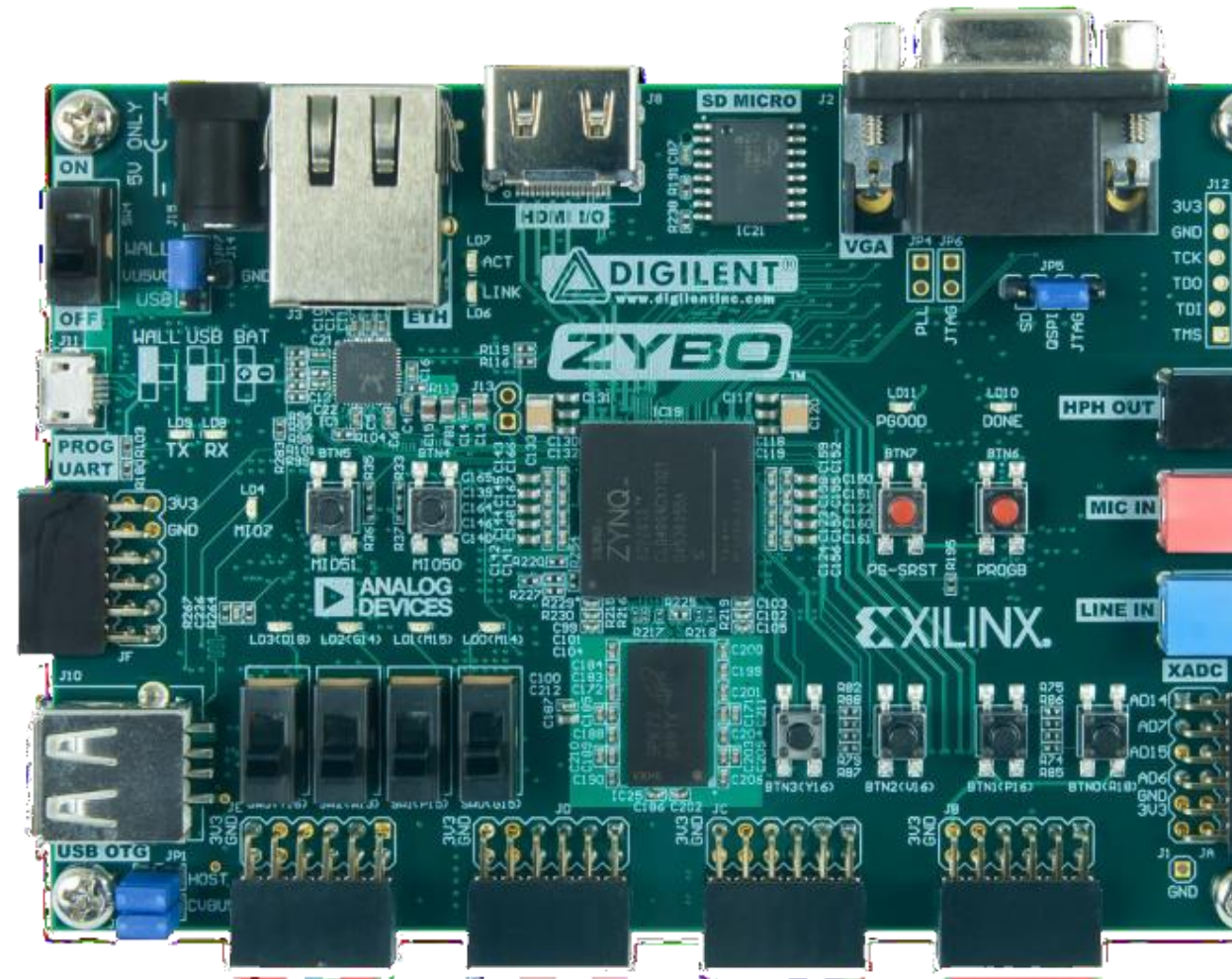
Digital Design Technology Map:

From a $f(x)$ to *Hardware Acceleration*



Hands-on:

VHDL modelling, logic synthesis and FPGA Programming



During the Lab Hands-On we will design from scratch a complete digital system, from requirements to the implementation on a FPGA Board and its test.

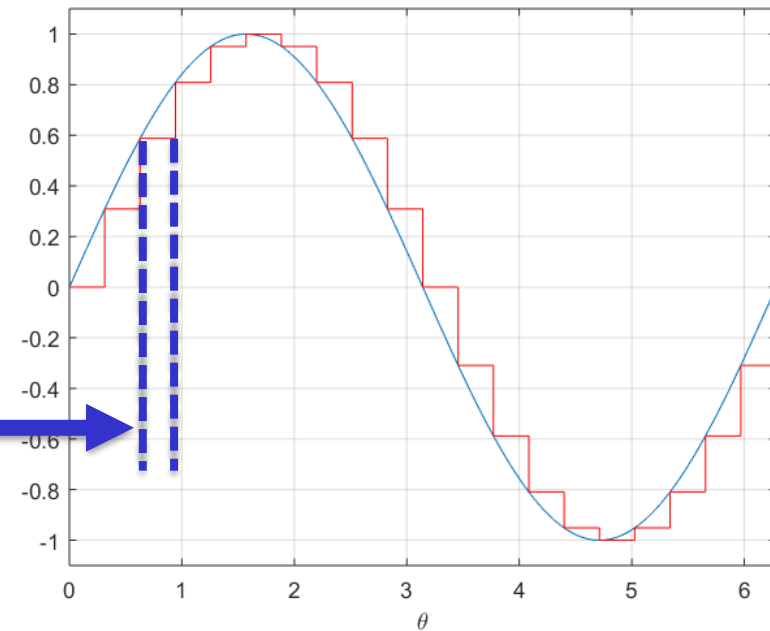
Hands-on: DDFS System Description

The DDFS (Direct Digital Frequency Synthesizer) is a digital system able to generate a quantized *sin* waveform, with frequency control.

Building block of many Telecommunication systems

Suppose we want to generate a single tone at frequency f :

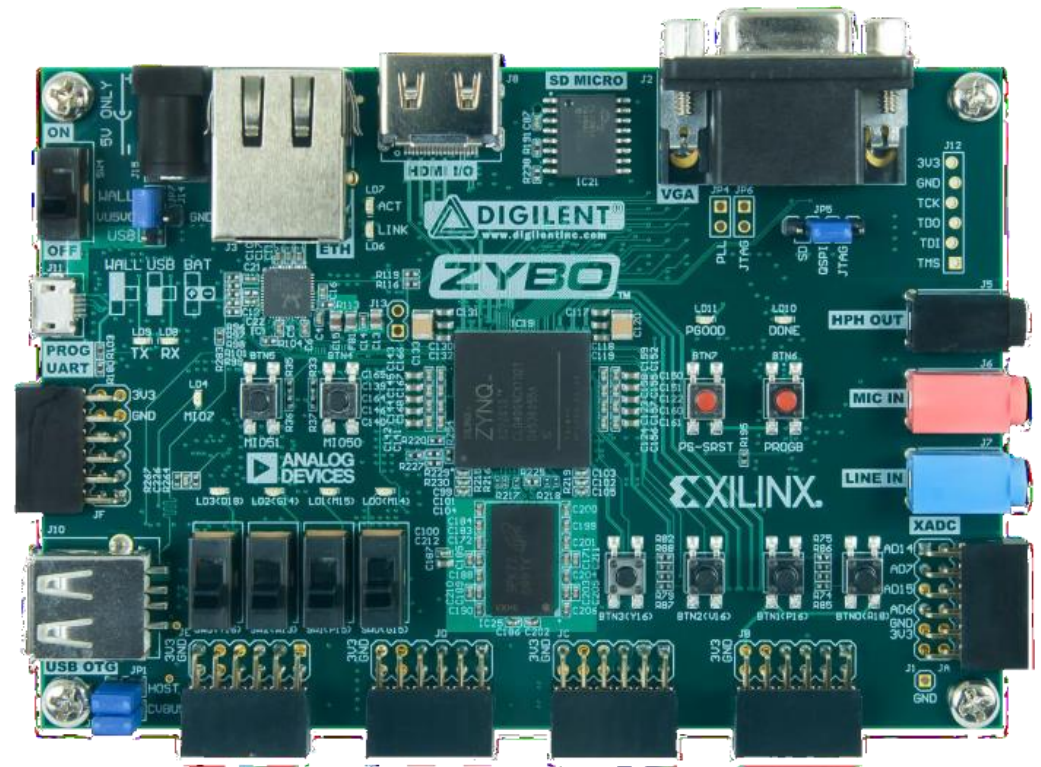
- Ideal output: $\sin(2\pi f t) = \sin(\theta)$
- Discrete time: $\sin(2\pi f k T_{ck}) = \sin(k\Delta\theta)$
- Phase resolution:
 $\Delta\theta = 2\pi \frac{f}{f_{ck}}$
 - Depends on f
 - Depends on f_{ck}



ZyBo Development Board (1/3)

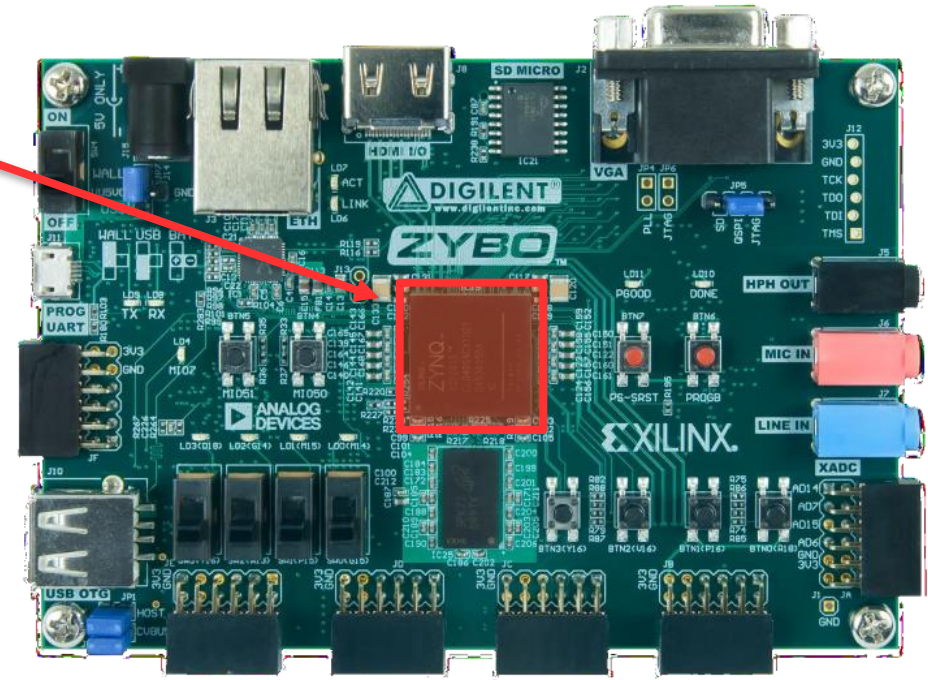
The ZYBO (ZYnq **BO**ard) is a feature-rich, ready-to-use, entry-level embedded software and digital circuit development platform built around the smallest member of the Xilinx Zynq-7000 family, the Z-7010.

The Z-7010 is based on the Xilinx All Programmable System-on-Chip (AP SoC) architecture, which tightly integrates a dual-core ARM Cortex-A9 processor with Xilinx 7-series Field Programmable Gate Array (FPGA) logic.



ZyBo Development Board (2/3)

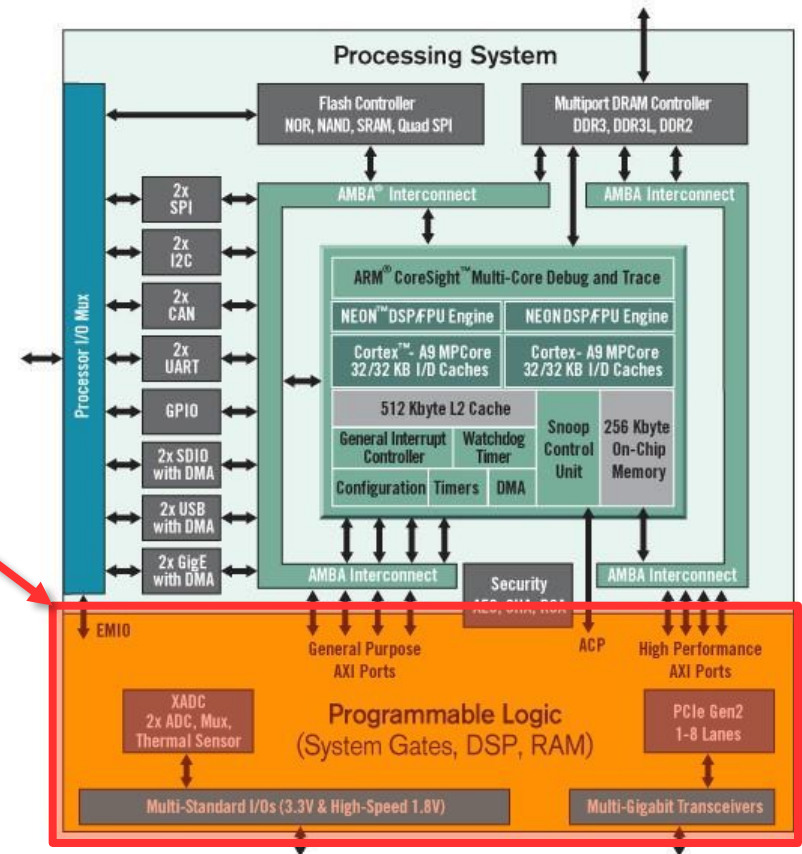
- **ZYNQ XC7Z010-1CLG400C**
- 512MB x32 DDR3 w/ 1050Mbps bandwidth
- Dual-role (Source/Sink) HDMI port
- 16-bits per pixel VGA source port
- Trimode (1Gbit/100Mbit/10Mbit) Ethernet PHY
- MicroSD slot (supports Linux file system)
- OTG USB 2.0 PHY (supports host and device)
- External EEPROM (programmed with 48-bit globally unique EUI-48/64™ compatible identifier)
- Audio codec with headphone out, microphone and line in jacks
- 128Mb Serial Flash w/ QSPI interface
- On-board JTAG programming and UART to USB converter
- GPIO: 6 pushbuttons, 4 slide switches, 5 LEDs
- Six Pmod connectors (1 processor-dedicated, 1 dual analog/digital, 3 high-speed differential, 1 logic-dedicated)



ZyBo Development Board (3/3)

ZYNQ XC7Z010-1CLG400C

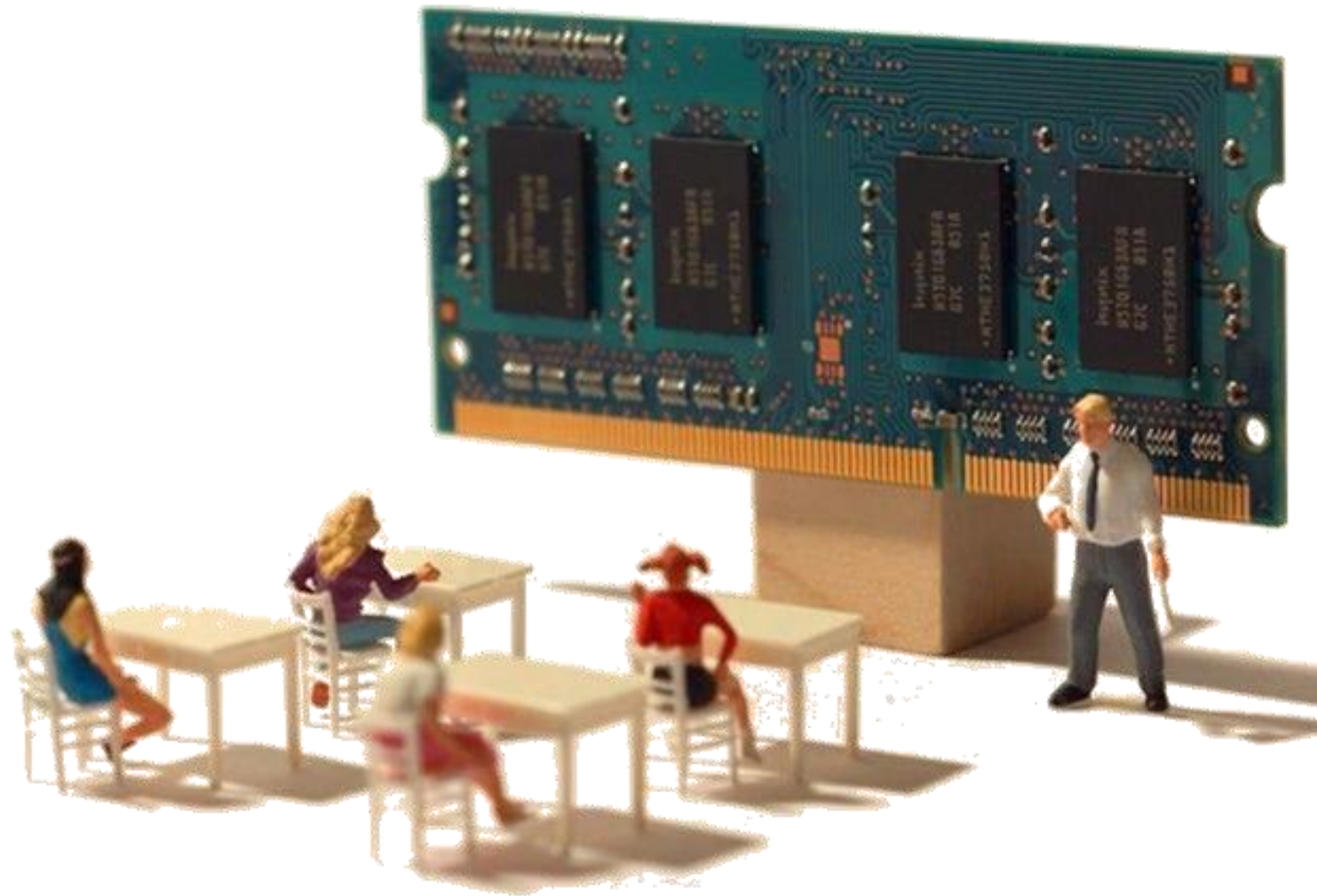
- 650Mhz dual-core Cortex-A9 processor
- DDR3 memory controller with 8 DMA channels
- High-bandwidth peripheral controllers: 1G Ethernet, USB 2.0, SDIO
- Low-bandwidth peripheral controller: SPI, UART, CAN, I2C
- **Reprogrammable logic equivalent to Artix-7 FPGA**
 - 4,400 logic slices, each with four 6-input LUTs and 8 flip-flops
 - 240 KB of fast block RAM
 - Two clock management tiles, each with a phase-locked loop (PLL) and mixed-mode clock manager (MMCM)
 - 80 DSP slices
 - Internal clock speeds exceeding 450MHz
 - On-chip analog-to-digital converter (XADC)



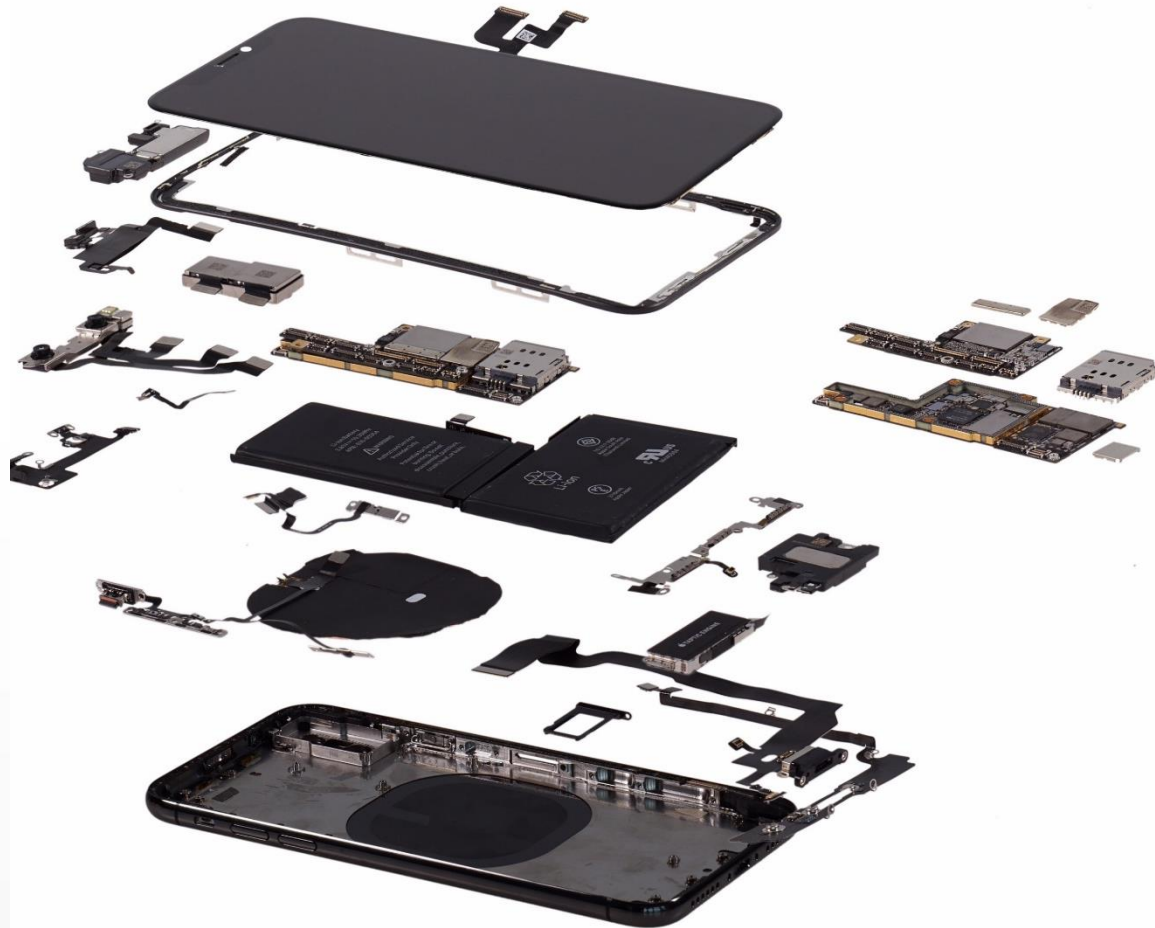
Exam

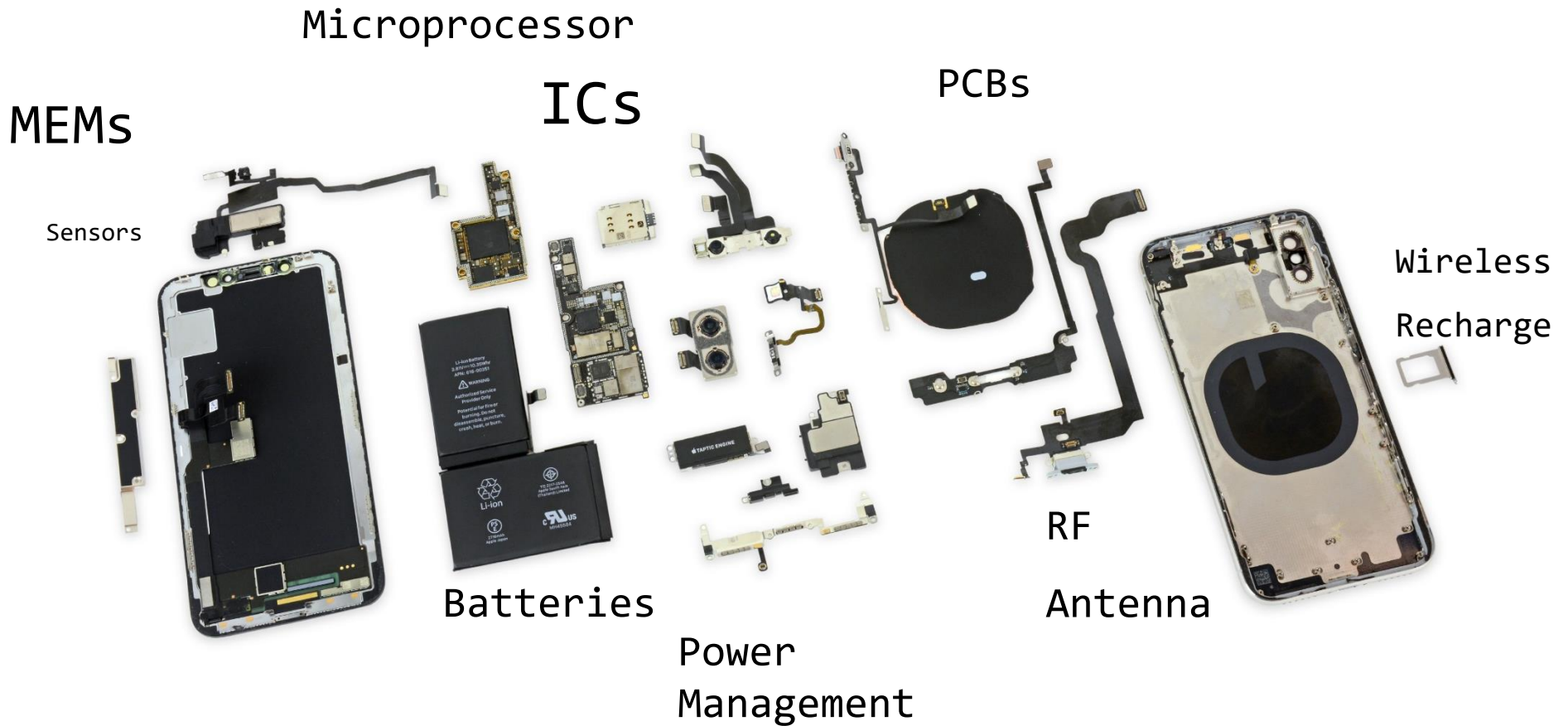
- Oral test on topics covered in the course
- Digital Design (VHDL and automatic logic synthesis for FPGA technology) -- (**optional BUT recommended**)
 - Realization of digital electronic system of similar complexity as seen during class hands-on exercises
 - 1 - 2 people
 - Request for project assignment via email
 - Email delivery of source code and testbench together with the technical report
 - Design Review/Discussion before the oral exam

Questions ?



Let's start !!



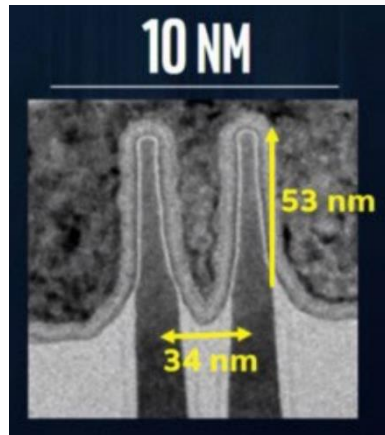
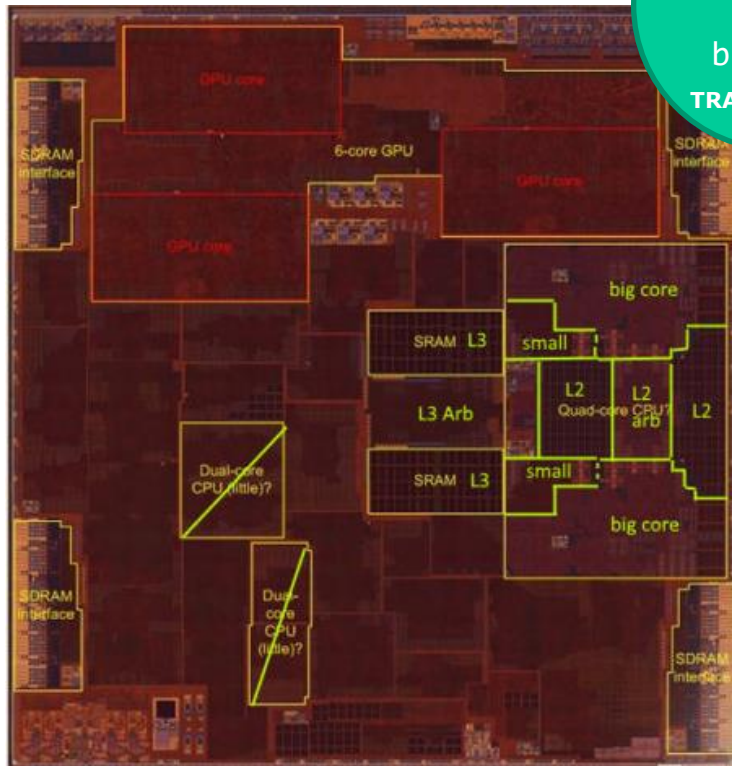


MULTIDISCIPLINARITY

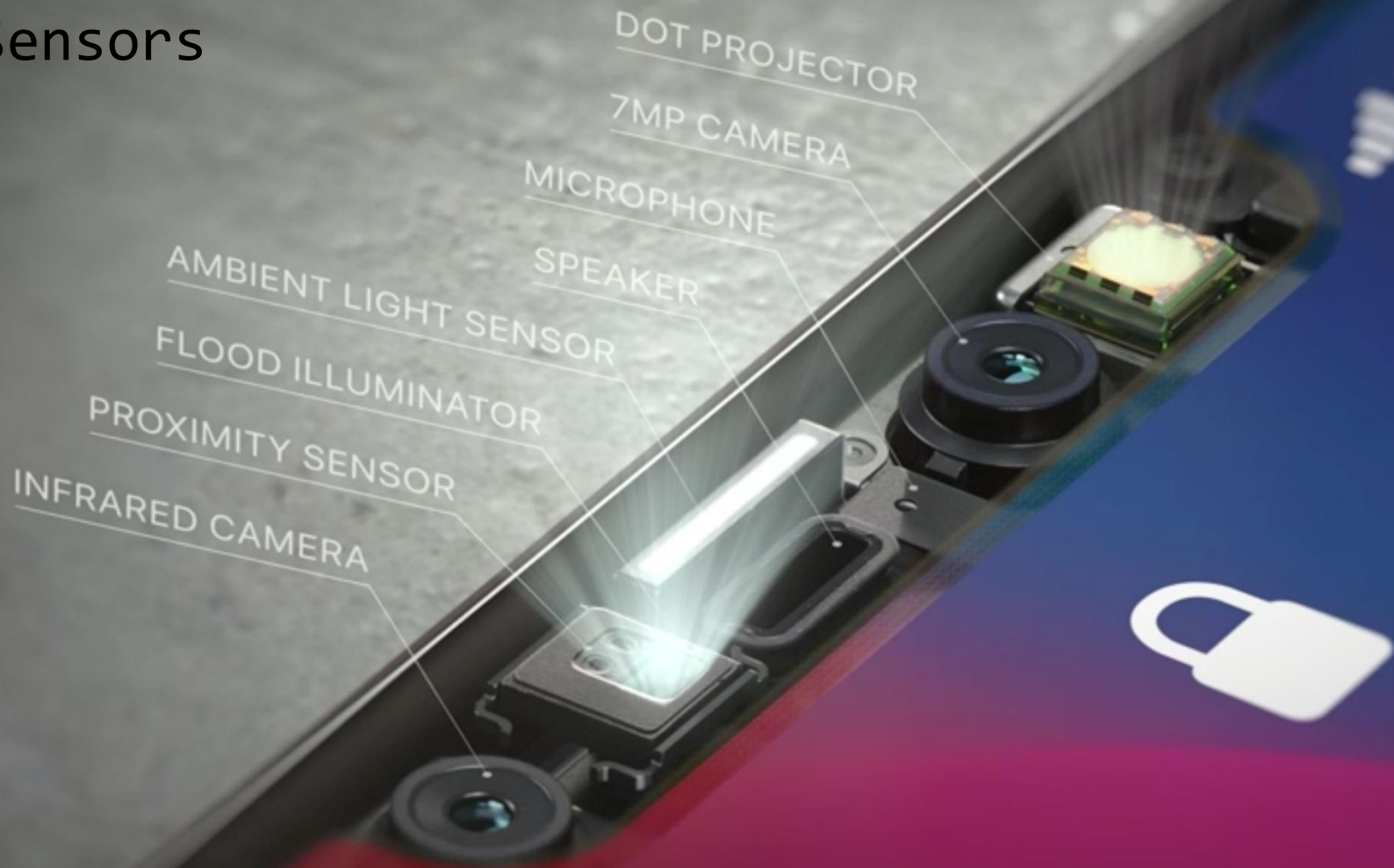
MicroProcessor

4.3

billion
TRANSISTORS

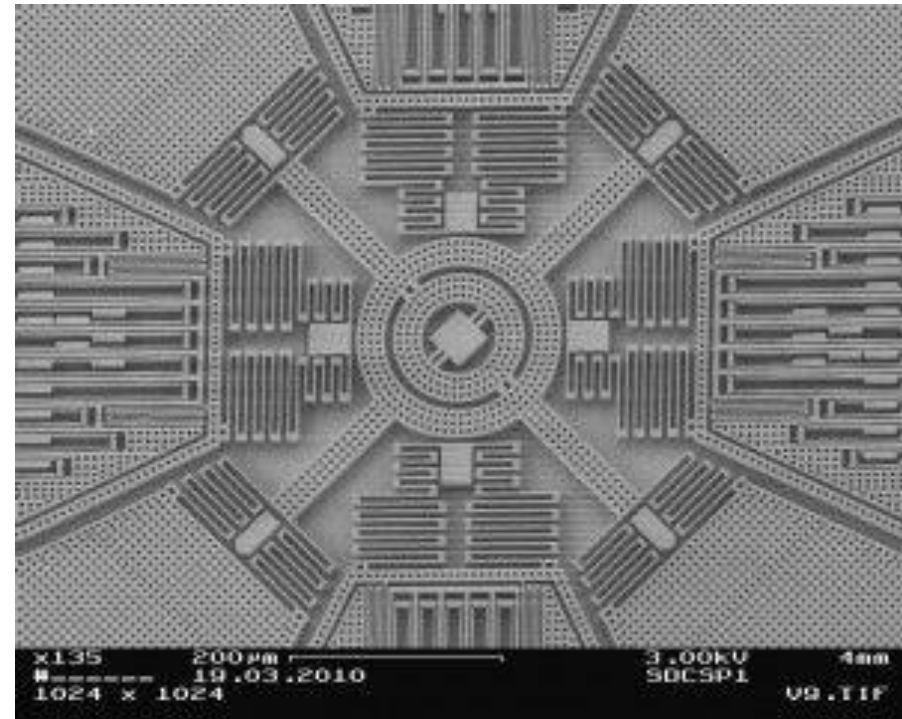


Sensors



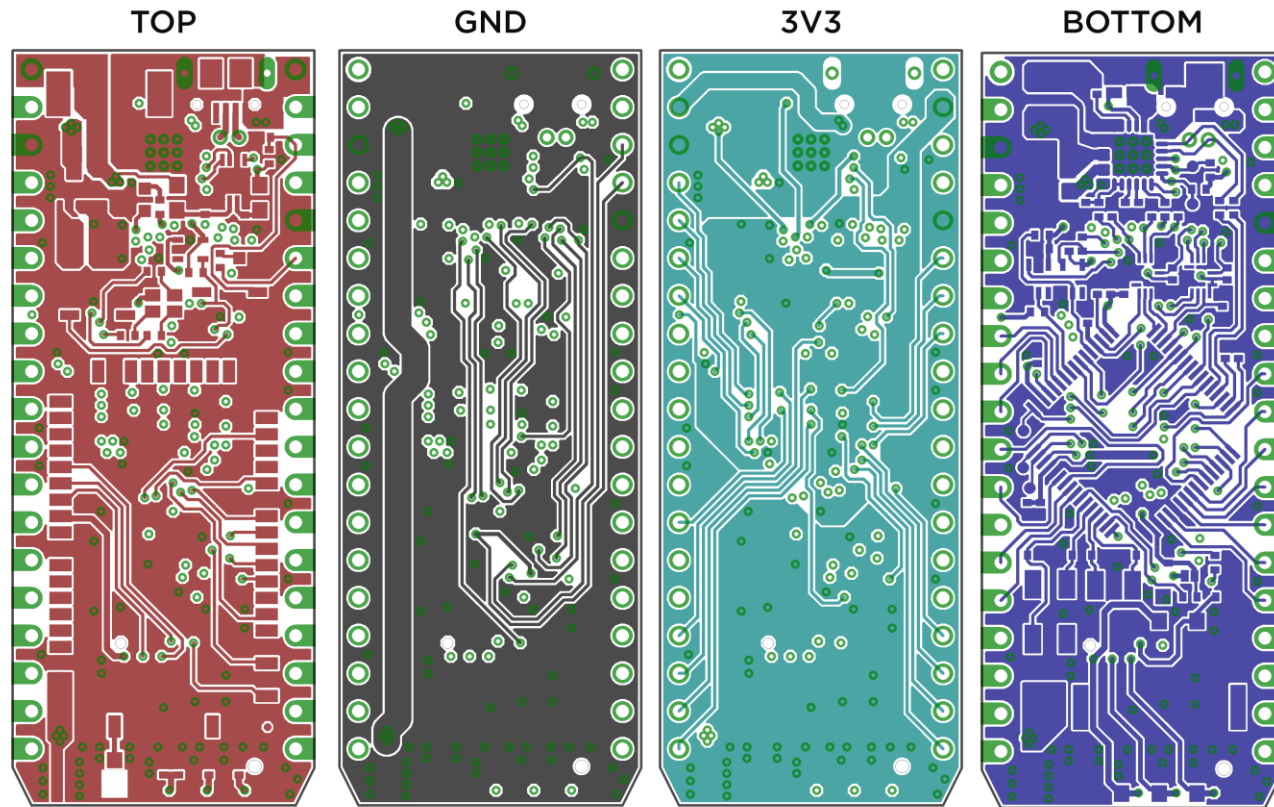
MEMs

Micro-Electro-Mechanical systems



PCB

Printed Circuit Boards



Batteries



Antenna





iPhone 8 Teardown

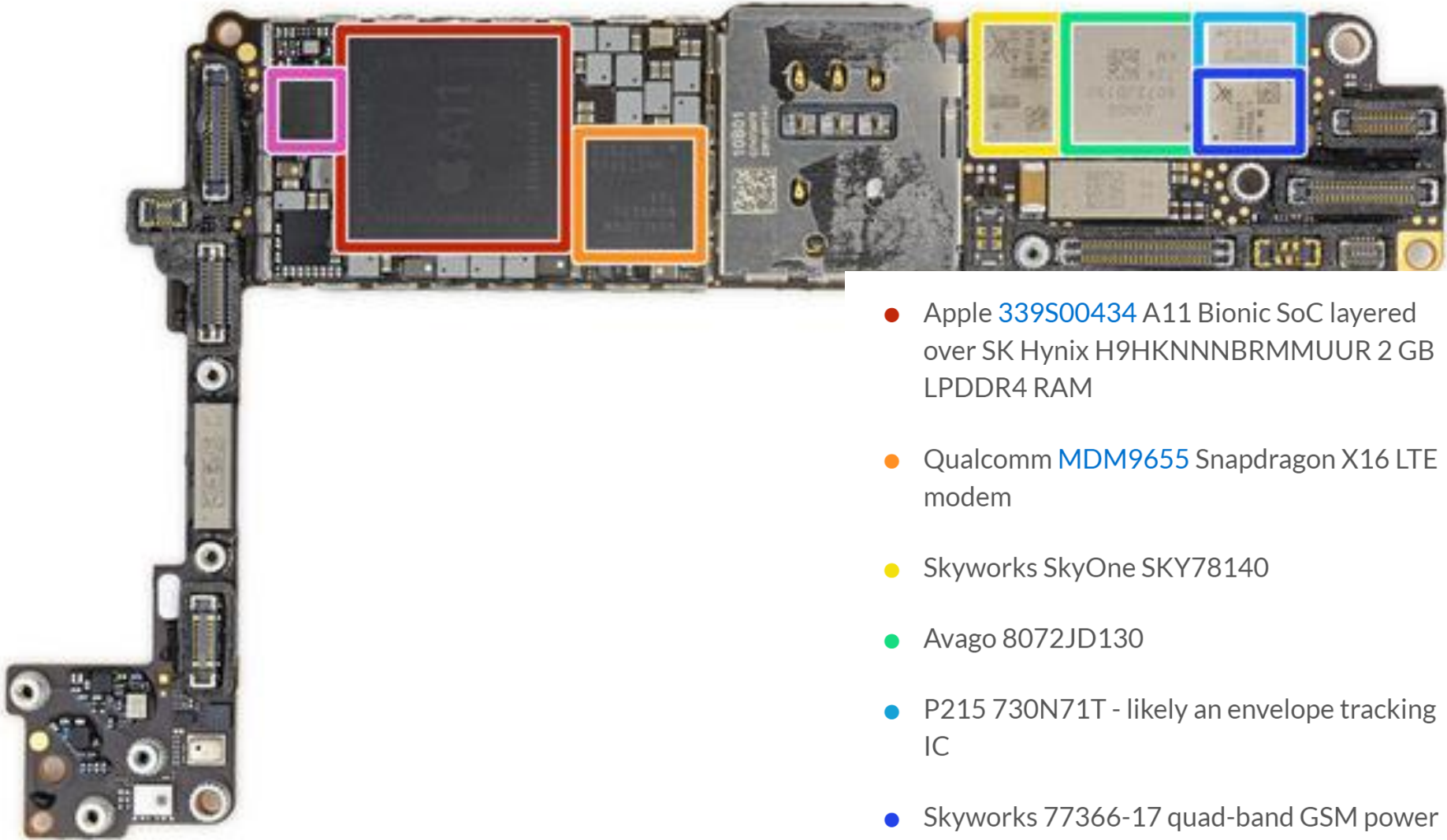
Published: September 21, 2017

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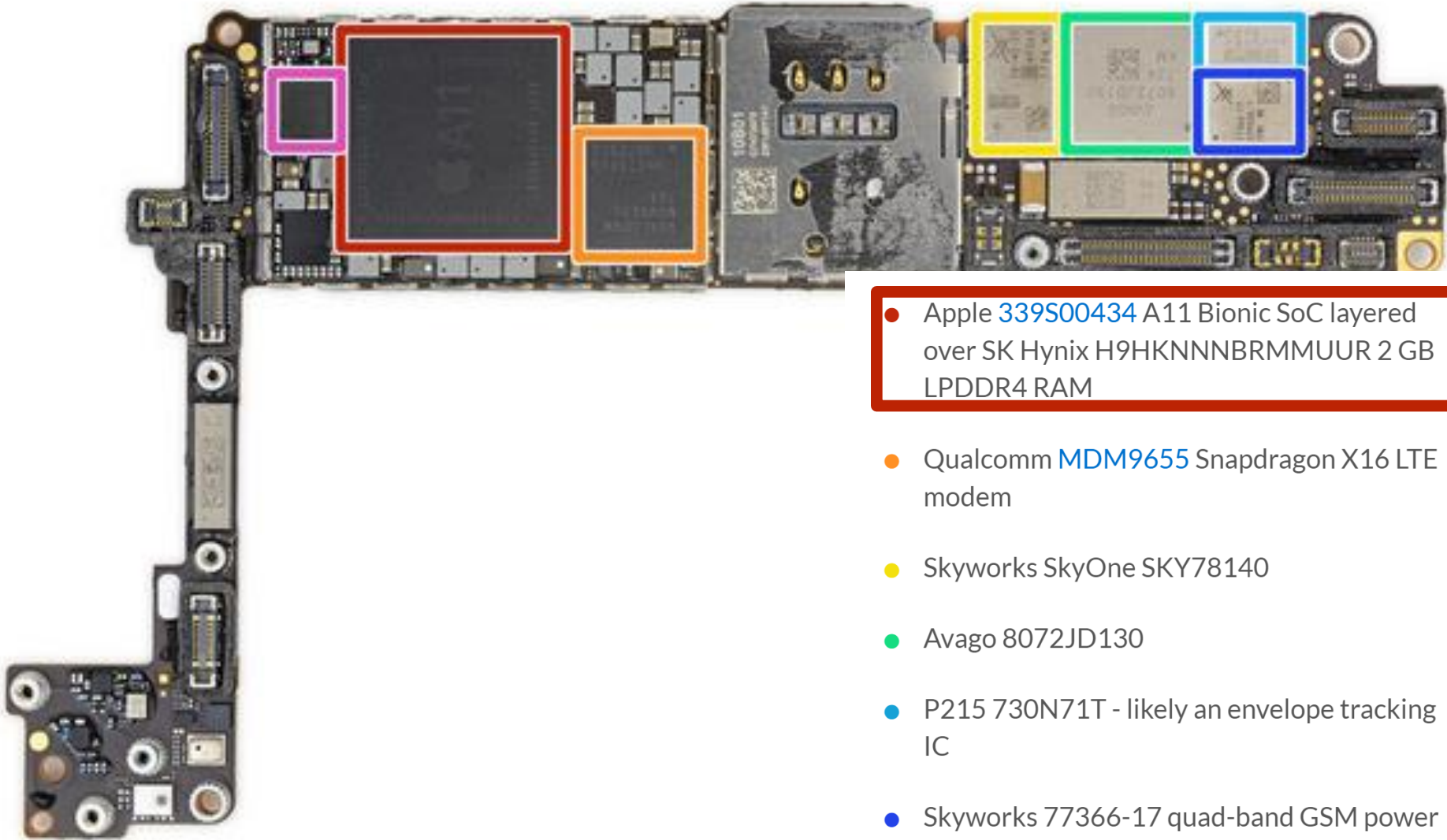
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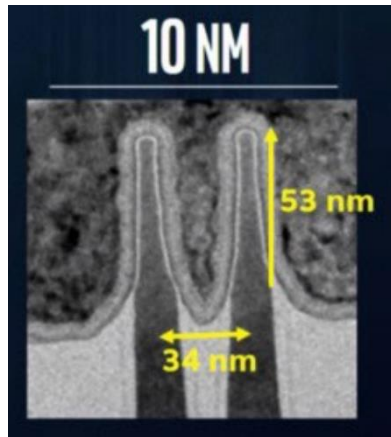


- Apple [339S00434](#) A11 Bionic SoC layered over SK Hynix H9HKNNNBRMMUUR 2 GB LPDDR4 RAM
- Qualcomm [MDM9655](#) Snapdragon X16 LTE modem
- Skyworks SkyOne SKY78140
- Avago 8072JD130
- P215 730N71T - likely an envelope tracking IC
- Skyworks 77366-17 quad-band GSM power amplifier module
- NXP [80V18](#) secure NFC module



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Taiwan Semiconductor Manufacturing Company



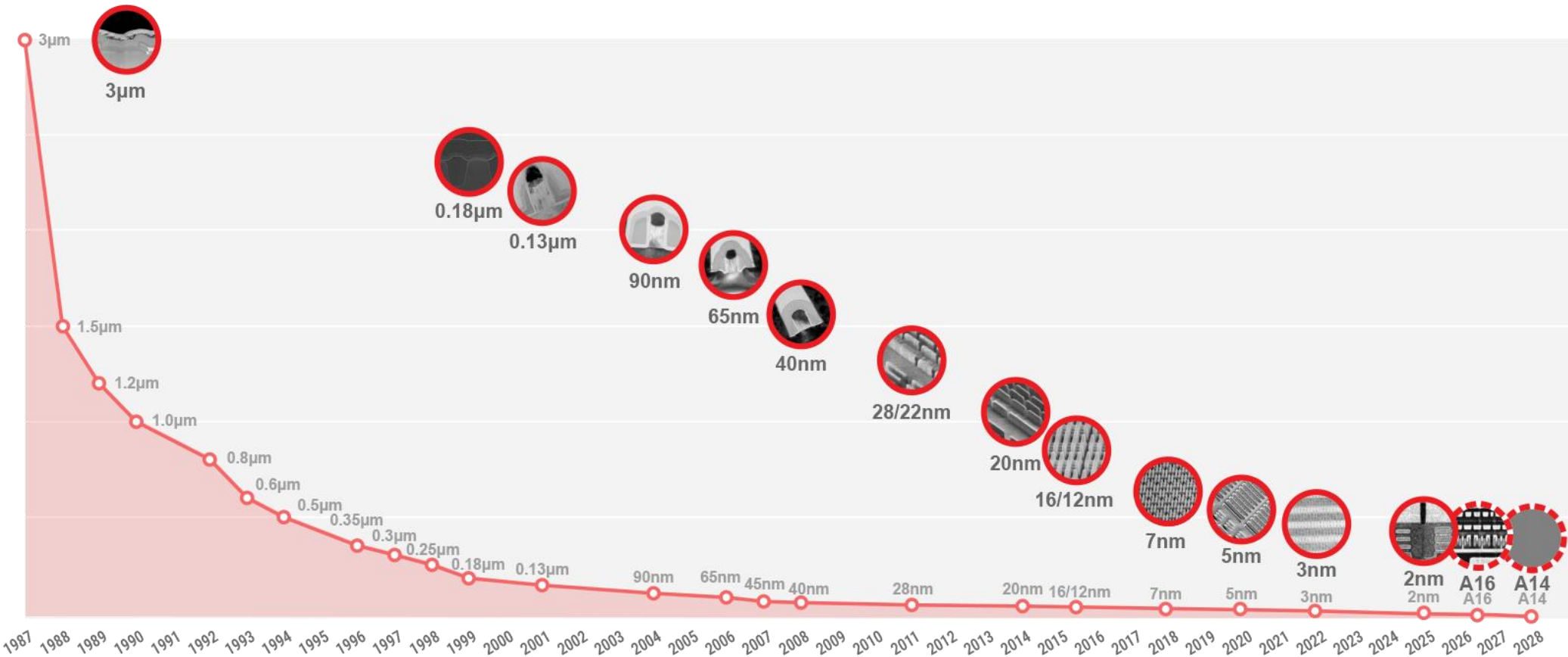
4.3
billion
TRANSISTORS



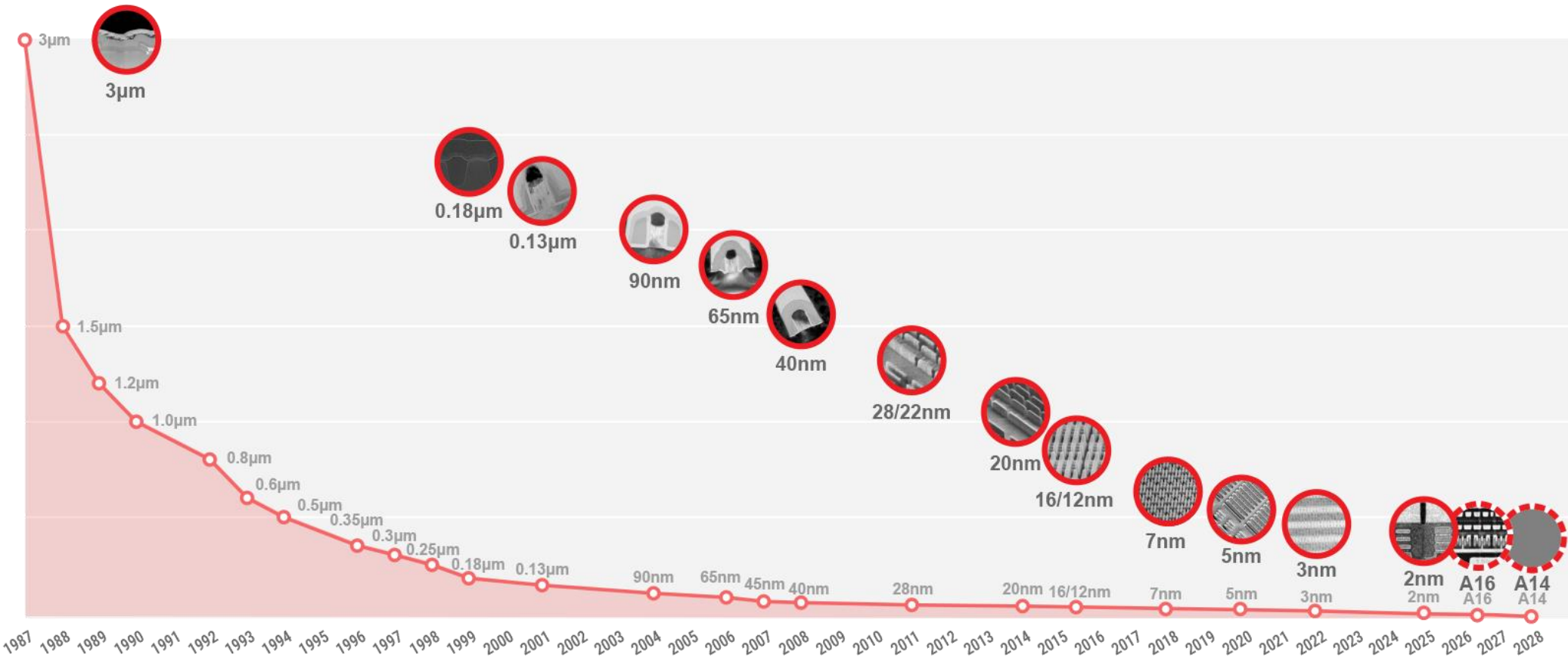
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- NXP [80V18](#) secure NFC module

iPhone 8

TSMC Logic Technology

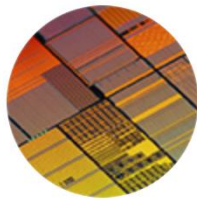


TSMC Logic Technology



A14 Technology

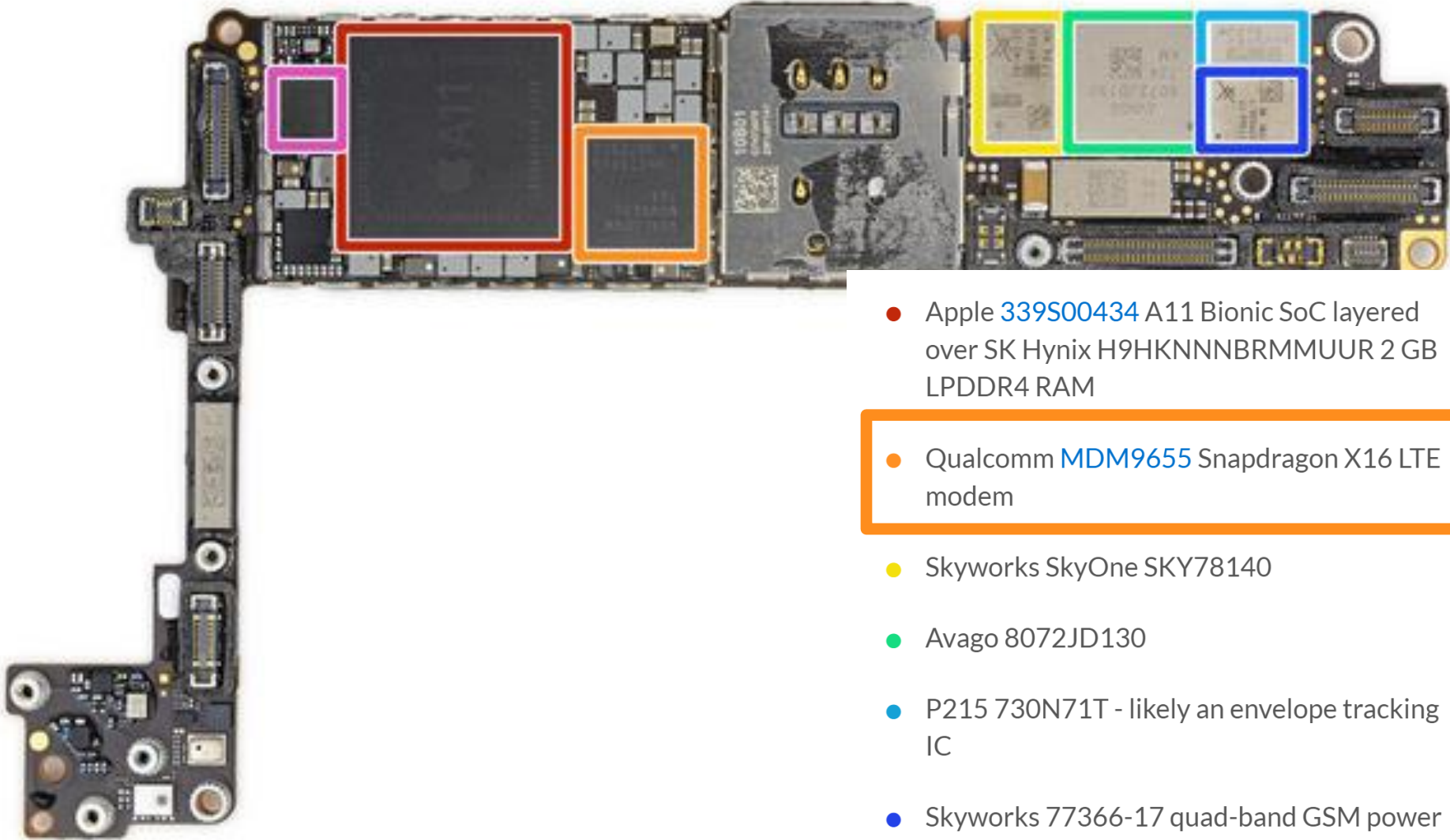
A14 technology is TSMC's next cutting-edge logic process that achieves full-node power, performance and area through dimensional scaling. A14 is designed to drive AI transformation forward by delivering...



A16 Technology

TSMC A16™ technology integrates leading nanosheet transistors with innovative Super Power Rail (SPR) solutions, bringing greatly improved logic density and performance. SPR improves logic density and...





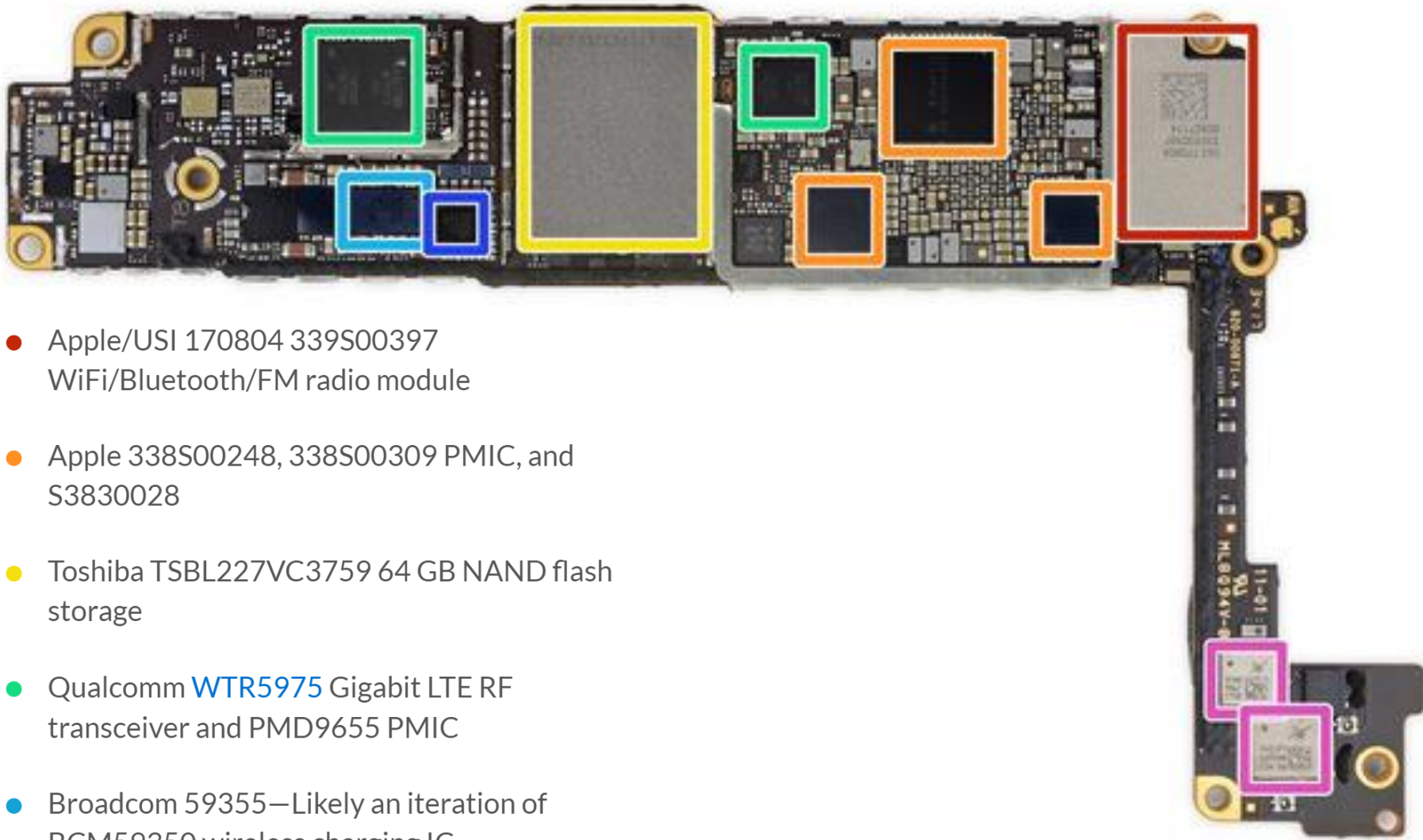
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<https://www.qualcomm.com/products/snapdragon-modems-4g-lte-x16>



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- Qualcomm [MDM9655](#) Snapdragon X16 LTE modem
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- P215 730N71T - likely an envelope tracking IC
- Skyworks 77366-17 quad-band GSM power amplifier module
- NXP [80V18](#) secure NFC module

iPhone 8



- Apple/USI 170804 339S00397
WiFi/Bluetooth/FM radio module
- Apple 338S00248, 338S00309 PMIC, and
S3830028
- Toshiba TSBL227VC3759 64 GB NAND flash
storage
- Qualcomm [WTR5975](#) Gigabit LTE RF
transceiver and PMD9655 PMIC
- Broadcom 59355—Likely an iteration of
BCM59350 wireless charging IC
- NXP 1612A1—Likely an iteration of the 1610
tristar IC
- Skyworks 3760 3576 1732 RF Switch and
SKY762-21 247296 1734 RF Switch

iPhone 8

Apple iPhone 8+ (A1864)

Preliminary Cost Summary

Total Cost Estimate 8+

Cost Of Manufacturing

\$288.08

\$7.36

Total Cost Estimate 7+

Cost Of Manufacturing

\$270.88

\$6.78

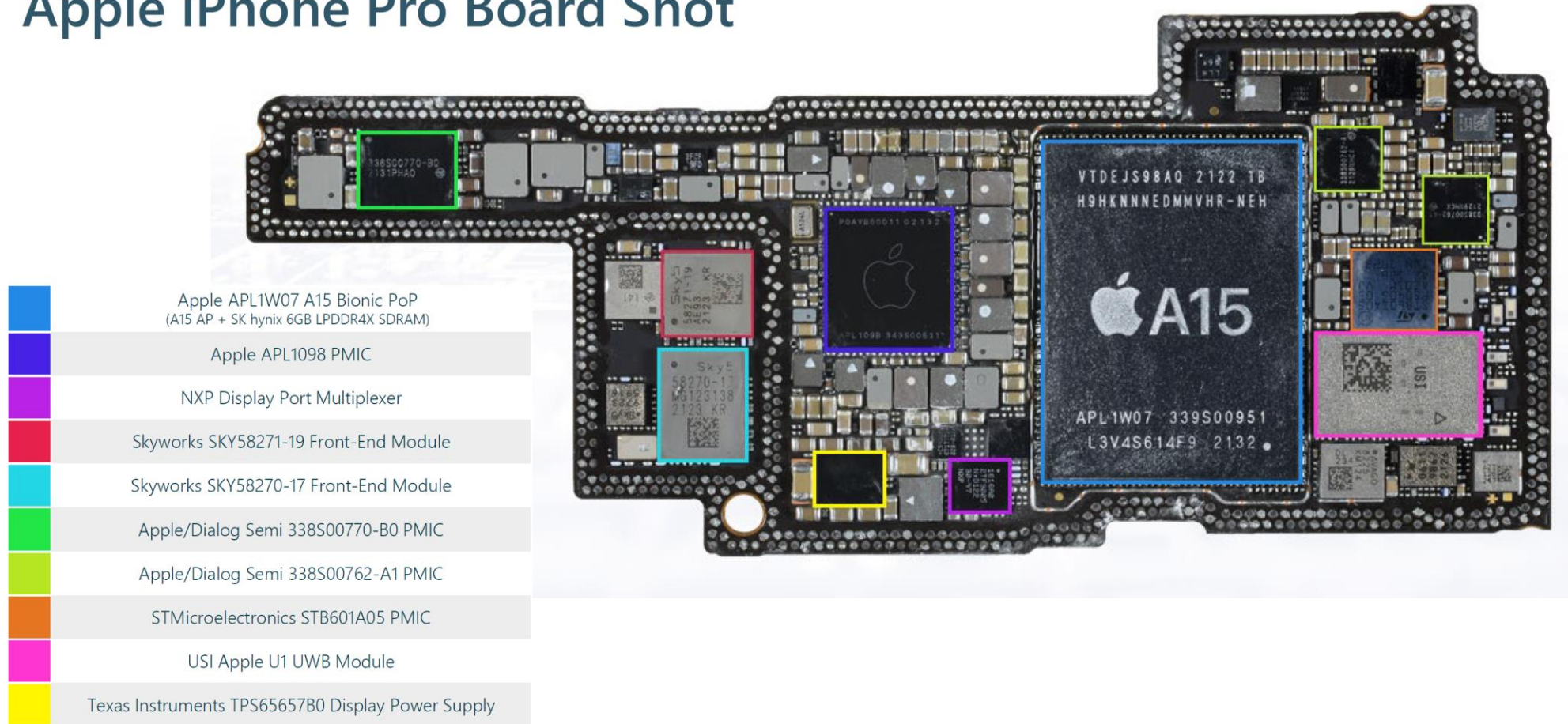
<< iPhone 7 Plus Finalized

Itemized Components	MfgName	MfgPartNbr	Description	Total Cost	Comment
Display	JDI, LGD, Sharp		5.5" 1920x1080 IPS LCD w/ in-cell touch	\$52.5000	See comments tab
Mechanical / Electro-Mechanical Components				\$50.9500	
Chassis + Rear Enclosure			Enclosure, Main, Bottom, Machined 7000-Series Aluminum Alloy w/stainless insert and Gorilla glass rear cover		
Other Mechanical / Electro-Mechanical			PCBAs, Connectors, Taptic Engine, Other		
Cameras				\$32.5000	
Primary Camera Module			Dual, Wide-Angle F1.8 / Telephoto F2.8 12MP w/ OIS		1.2umx1.2um Pixel Size Wide Angle; 1.0x1.0um Pixel Size Telephoto
Secondary Camera Module			7MP F2.2		1.0umx1.0um Pixel Size
Memory				\$31.2000	
NAND (eMMC, MLC, ...)	SANDISK CORP	SDMPEGFI2 064G	Flash, NAND, 64GB, TLC		Samsung, Toshiba, SanDisk, Hynix are all vendors
DRAM	MICRON TECHNOLOGY INC	MT53D384M64D4NY-046 XT:D	SDRAM, LPDDR4, 3GB, PoP		
Apps Processor	Apple		Apple A11 Bionic, 64-Bit 6-Core CPU, 3- Core GPU, 10nm	\$27.5000	

Itemized Components	MfgName	MfgPartNbr	Description	Total Cost	Comment
RF / PA Section				\$24.6000	
RF Transceiver	QUALCOMM	WTR5975	RF Transceiver, Multi-Mode, Multi-Band		
Transmit Modules	BROADCOM LTD	AFEM-8072	Transmit Module		
	SKYWORKS SOLUTIONS INC	SKY78140-22	Transmit Module		
PAMs	SKYWORKS SOLUTIONS INC	SKY77366-17	Transmit Module		
Other RF/PA			Remaining PAMs ad other RF components not listed above		
Power Management Components				\$16.0500	
Apps Processor Power Management	DIALOG SEMICONDUCTOR GMBH		Power Management IC		
Baseband & Other Power Management	QUALCOMM	PMD9655	Power Management IC		
Wireless Charging Element + Supporting Compor	BROADCOM LTD	BCM59355A2IUB3G	Wireless Charging IC		
Other Power Management Components			Remaining power management ics, regulators, converters, LED drivers, etc.		
Baseband IC				\$11.5000	
	QUALCOMM	MDM9655	Baseband Processor, Multi-Mode, 14nm - CAT16 Modem		
User Interface Components				\$11.2800	
NFC Controller	NXP SEMICONDUCTORS	PN80V	NFC Controller		
Audio Codecs	CIRRUS LOGIC INC	338S00248	Audio Codec		
	CIRRUS LOGIC INC	CS35L26	Audio Power Amplifier		
Other			Includes camera flash, and other interface ics		
Other Modules					
WLAN / BT Module(s)	MURATA	339S00399	BT / WLAN Module, IEEE802.11ac, Bluetooth 5.0	\$7.3500	
Sensors				\$6.6500	
Fingerprint Sensor Module			Fingerprint Sensor Module		
Accelerometer / Gyroscope, 6-Axis			Accelerometer / Gyroscope, 6-Axis		
Electronic Compass	ALPS ELECTRIC CO LTD		Electronic Compass		
Barometric Pressure Sensor	BOSCH SENSORTec GMBH	BMP28x	Barometric Pressure Sensor		
Color Sensor	AMS AG		Color Sensor		
Proximity - time-of-flight			Proximity - time-of-flight		
Battery Pack(s)				\$4.4500	
	HUIZHOU DESAY BATTERY CO LTD		Battery, Li-Polymer, 3.82V, 2691mAh, 10.28Wh		
Box Contents				\$11.5500	
Charger, 5V, 1A, AC to USB Type A			Includes pkg & literature Charger, 5V, 1A, AC to USB Type A		
Stereo w/ Apple Lightning Plug			Stereo w/ Apple Lightning Plug		
Headphone Jack Adapter			Headphone Jack Adapter		
USB to Apple Lightning Cable			USB to Apple Lightning Cable		

iPhone 13

Apple iPhone Pro Board Shot



iPhone 13

Apple iPhone Pro Board Shot

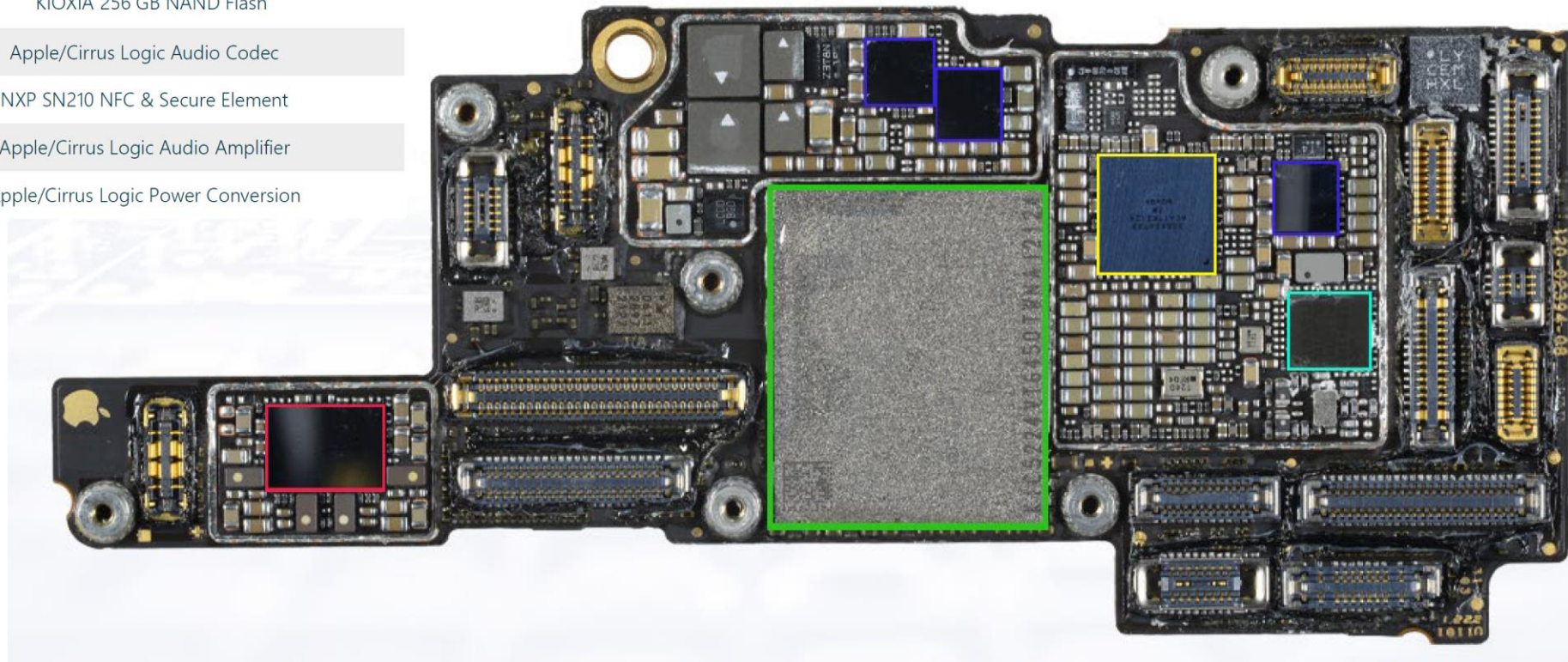
KIOXIA 256 GB NAND Flash

Apple/Cirrus Logic Audio Codec

NXP SN210 NFC & Secure Element

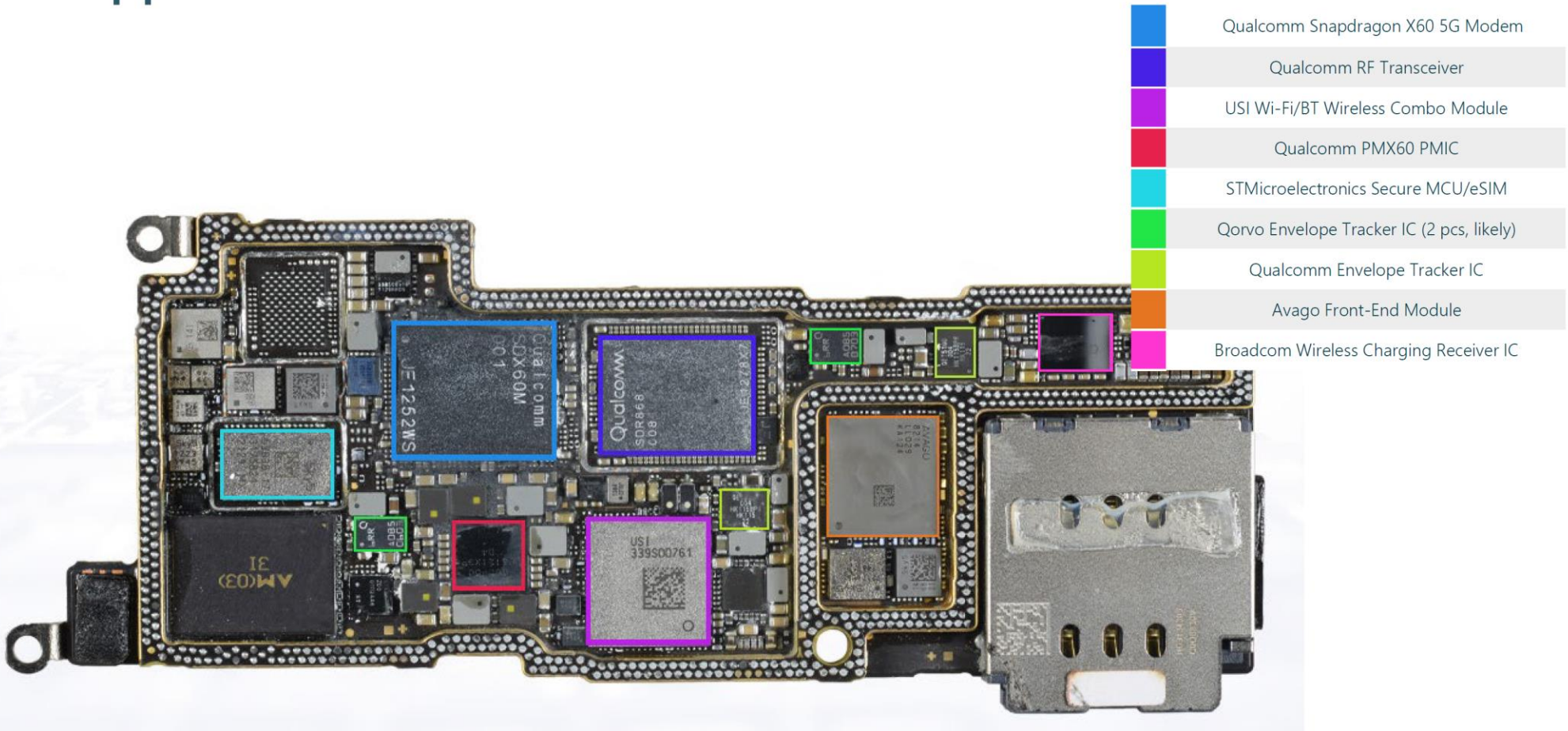
Apple/Cirrus Logic Audio Amplifier

Apple/Cirrus Logic Power Conversion

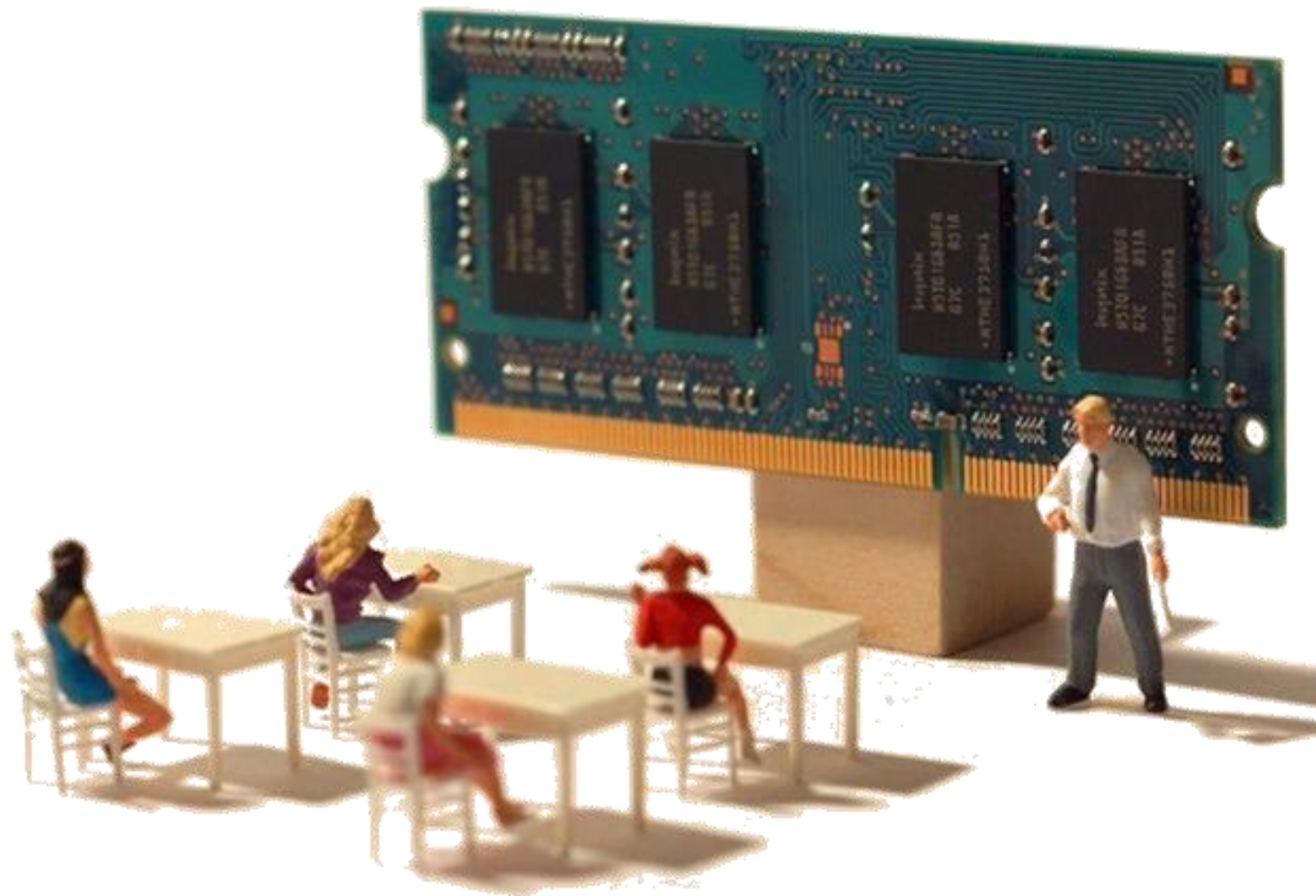


iPhone 13

Apple iPhone Pro Board Shot



Questions ?



Some Companies in this market



Jobs' Profiles (some examples)

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<http://jobs.st.com/>

<http://www.qualcomm.com/careers>

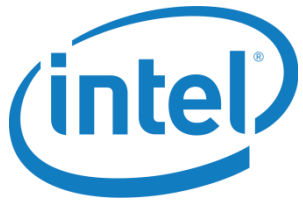
<http://www.imgtec.com/corporate/careers.asp>

<https://jobs.intel.com/page/show/careers>

<http://www.infineon.com/cms/en/corporate/career/infineon-jobsearch/>

<http://www.dialog-semiconductor.com/careers/>

<http://arm.com/careers/>



<https://www.apple.com/jobs/us/index.html>



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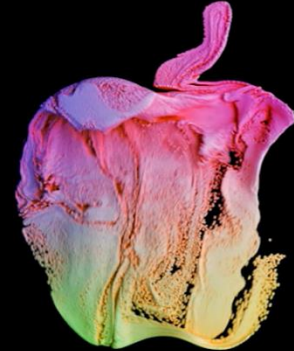
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Digital Design Engineer (m/f/d)

Livorno

Hardware

Dec 17, 2024

PMU Analog Design Intern

Livorno

Hardware

Sep 11, 2024

PMU RTL/ Digital Design Intern

Livorno

Hardware

Sep 4, 2024

Digital Design Engineer (m/f/d)

Livorno, Livorno, Italy

Hardware



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Summary

Posted: **Dec 17, 2024**

Role Number: **200582651**

At Apple, we work every single day to craft products that enrich people's lives. The people here at Apple don't just build products - they create the kind of wonder that revolutionizes entire industries. We invite you to join our dynamic group, for the unique and rewarding opportunity to contribute to upcoming products that will delight and inspire millions of Apple's customers every day! Apple's PMU Hardware Tech team are responsible for delivering the power in a highly configurable and controlled way to the high end Apple SoCs, which power everything from Apple Watch and Apple TV to iPhone, iPad and Mac. We're looking for dedicated Engineers who can help us to innovate the way we verify power management devices, to provide industry leading power and battery efficiency and achieve customer expectations of device performance. Do you love working on challenges that no one has solved yet? Then we welcome you to work among the industry's best!

As a member of the Digital Design Team, you will assist with delivering high quality chips in order to meet performance, feature, timing, area, power and efficiency goals. We will provide you with the mentorship and the opportunity to collaborate with world class Design Engineers, as well as work alongside our design verification, SoC platform architecture, and physical design teams. You will get the opportunity to be part of a team delivering PMU silicon directly into next generation Apple products.

Description

You will work within the Digital Design Team to refine or create requirements and specifications, collaborate with the analog design and design verification teams, create/update RTL designs, and run simulations to check your design. This could include tasks such as implementing bespoke control of analog circuits, creating state machines to control the system, or working on industry standard interfaces to high performance SoCs. Your designs will need to balance energy efficiency and area constraints with project schedule and maintainability. You will also review synthesis and power reports, root-cause and resolve timing and power issues and ensure maximal QoR throughout your design.

Minimum Qualifications

- A Master's or Ph.D. in electrical engineering, Communication Engineering, Computer Science/ Software Engineering, or equivalent.
 - Apple is an Equal Opportunity Employer that is committed to inclusion and diversity. We also take affirmative action too offer employment and advancement opportunities to all applicants, including minorities, women, protected veterans, and individuals with disabilities.
-

Preferred Qualifications

- Strong familiarity with modern, energy-efficient/low-power logic design techniques.
- Strong familiarity with RTL design in Verilog/VHDL, CPU bus architectures, and mixed signal design.
- Strong familiarity with synthesis, STA and quality checks such as Lint and CDC/RDC finite state machines.
- Excellent interpersonal skills and well-organized working style. Ability to work well in a team and be productive under tight schedules.
- Strong analytical/problem solving skills.
- Proficient in using (System)Verilog, the ability to analyze RTL/Netlist designs, and outstanding debugging skills to tackle technical challenges.

PMU RTL/ Digital Design Intern

Livorno, Livorno, Italy

Hardware



Submit Resume

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Summary

Posted: **Sep 4, 2024**

Role Number: **200566308**

At Apple, we work every single day to craft products that enrich people's lives. The people here at Apple don't just build products – they create the kind of wonder that revolutionizes entire industries. We invite you to join our dynamic group, for the unique and rewarding opportunity to contribute to upcoming products that will delight and inspire millions of Apple's customers every day!

Apple's PMU Hardware Tech team are responsible for delivering the power in a highly configurable and controlled way to the high end Apple SoCs, which power everything from Apple Watch and Apple TV to iPhone, iPad and Mac.

We're looking for dedicated students who can help us to innovate the way we verify power management devices, to provide industry leading power and battery efficiency and achieve customer expectations of device performance. Do you love working on challenges that no one has solved yet? Then we welcome you to work among the industry's best.

As a member of the Digital Design Team, you will assist with delivering high quality chips in order to meet performance, feature, timing, area, power and efficiency goals. We will provide you with the mentorship and the opportunity to collaborate with expert Design Engineers, as well as work alongside our design verification, SoC platform architecture, and physical design teams. You will get the opportunity to be part of a team delivering PMU silicon directly into next generation Apple products.

Description

With mentorship, you will work within the Digital Design Team to refine or create requirements and specifications, collaborate with the analog design and design verification teams, create/update RTL designs, and run simulations to check your design.

This could include tasks such as implementing bespoke control of analog circuits, creating state machines to control the system, or working on industry standard interfaces to high performance SoCs. Your designs will need to balance energy efficiency and area constraints with project schedule and maintainability. You may also review synthesis and power reports, root-cause and resolve timing and power issues and ensure maximal QoR throughout your design.

Minimum Qualifications

- Enrolled in Bachelors/Masters/PhD studies in EE or related field
- Excellent interpersonal skills and well-organised working style
- Fluent English skills are required
- Availability for at least 6 months

Preferred Qualifications

- Coursework focusing on modern, energy-efficient/low-power logic design techniques
- Strong familiarity with RTL design in Verilog/VHDL and understanding of the logic structures being inferred
- Ability to work well in a team and be productive under tight schedules
- Strong analytical/problem solving skills
- An understanding of finite state machines, CPU bus architectures, and mixed signal design

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$$P_a = P_i G_i C$$
$$(K_i R_i)$$
$$S_i = 4V_d$$
$$+ \left(\frac{I_0}{I_D} \right)$$
$$I_i = 7\% \uparrow$$



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$$S_i = 4V_d$$
$$+ \left(\frac{I_0}{I_D} \right)$$
$$I_i = 7\% \uparrow$$



Staff Digital Design Engineer

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ID: 3070750

Job Description

Job Posting Date	Additional Job Posting Location
2025-01-31	Bristol GBR

Company:
Qualcomm Technologies International Ltd

Job Area:
Engineering Group, Engineering Group > ASICS Engineering

General Summary:

Qualcomm invents breakthrough technologies that transform how the world computes, connects, and communicates. Today, our inventions are the foundation for life-changing products, experiences, and industries.

Qualcomm's Voice and Music Business Unit is a leading player in wireless earbud, headset, and smart speaker market. The business unit develops and delivers hardware, software and applications that bring together the very latest wireless and audio technologies to create industry leading audio voice and music products. You will be joining a successful engineering department whose deliveries can be found in billions of Bluetooth and Voice and Music products worldwide. We offer end-to-end solutions for leading-edge low-power mixed-signal SOCs from feasibility study through to system architecture, digital and analogue design, verification and validation, implementation and layout, embedded software design, ATE test, production, and characterisation.

Based in either our Cambridge or Bristol office, this role offers a Staff Engineer position in Digital Design working alongside System architects, Verification, Validation and Software teams.

What We are Looking For

- 5 - 10 years industry experience with Master's degree or PhD in Electrical, Electronic, Computer Engineering, or related field (or Bachelor's with equivalent experience).
- Experience of complex digital block/sub-system architecture development, design, verification and integration.
- A mixture of demonstrable technical skills and hands-on design experience in low power digital IP.
- An engineer who is detail-oriented with strong organizational, problem solving, communication and collaboration skills.
- A member of the team who is proactive, creative, curious, motivated to learn and contribute.

Key Responsibilities

- Ownership of complete work package from specification through to delivery.
- Design, integration and verification of RTL blocks including DSP, SoC and peripheral IP.
- Perform front-end design tasks including but not limited to: RTL coding, simulation, validation plan development, synthesis, timing closure, ECOs, etc.
- Planning and management of project work with predictable outcomes.
- Definition of high-level specifications and micro-architecture for digital low power IPs.
- Collaboration across global teams throughout the process of IP specification and use case definition, verification, bring-up and validation, etc.
- Running ASIC development tools including PLDRC, CDC, synthesis.
- Reporting status and communicating progress against expectations.
- Preparation of technical documentation.
- Understanding of customer use cases and specifying requirements.

Minimum Qualifications:

- Bachelor's degree in Science, Engineering, or related field and 4+ years of ASIC design, verification, validation, integration, or related work experience.

OR

Master's degree in Science, Engineering, or related field and 3+ years of ASIC design, verification, validation, integration, or related work experience.

OR

PhD in Science, Engineering, or related field and 2+ years of ASIC design, verification, validation, integration, or related work experience.

Required competencies

- Effective communicator with excellent technical presentation skills.
- Ability to work autonomously and demonstrate technical competency.
- Knowledge of DSP design for Audio or wireless modem technologies.
- Experience of SoC level design and IP Integration.
- RTL design in Verilog/System Verilog and expert digital design knowledge.
- Experienced with RTL and Gate-Level power analysis.
- Understanding of scalable design consideration.
- Experience of multiple clock domain crossings design.
- Understanding of multiple power domains design.
- Scripting skills on Linux, TCL and Perl/Python.
- Experience with EDA tools (eg. Cadence, Synopsys, Spyglass).
- Knowledge and experience of defining HW/FW interfaces.
- Expert in debugging design issues reported in simulation or formal verification.
- Understanding of coverage driven verification methodology.
- Familiarity with the Arm AMBA AXI/AHB specifications or other bus protocols.
- Ability to automate and improve process through scripting in Python, Perl.
- Demonstrates good understanding of, and conducts research on, industry trends and innovations in digital low power circuit design to ensure solutions and deliverables align with best practices.

Desirable competencies

- Familiarity with Audio DSP techniques.
- Familiarity with Bluetooth protocol and PHY.
- Familiarity with peripheral interfaces like I2C, SPI, UART, PWM.
- Familiarity with wireless communication standards (eg. UWB, WLAN, NFC).
- Experience of SystemC and High-Level Synthesis.
- Understanding of system modelling methodology using high level programming language.

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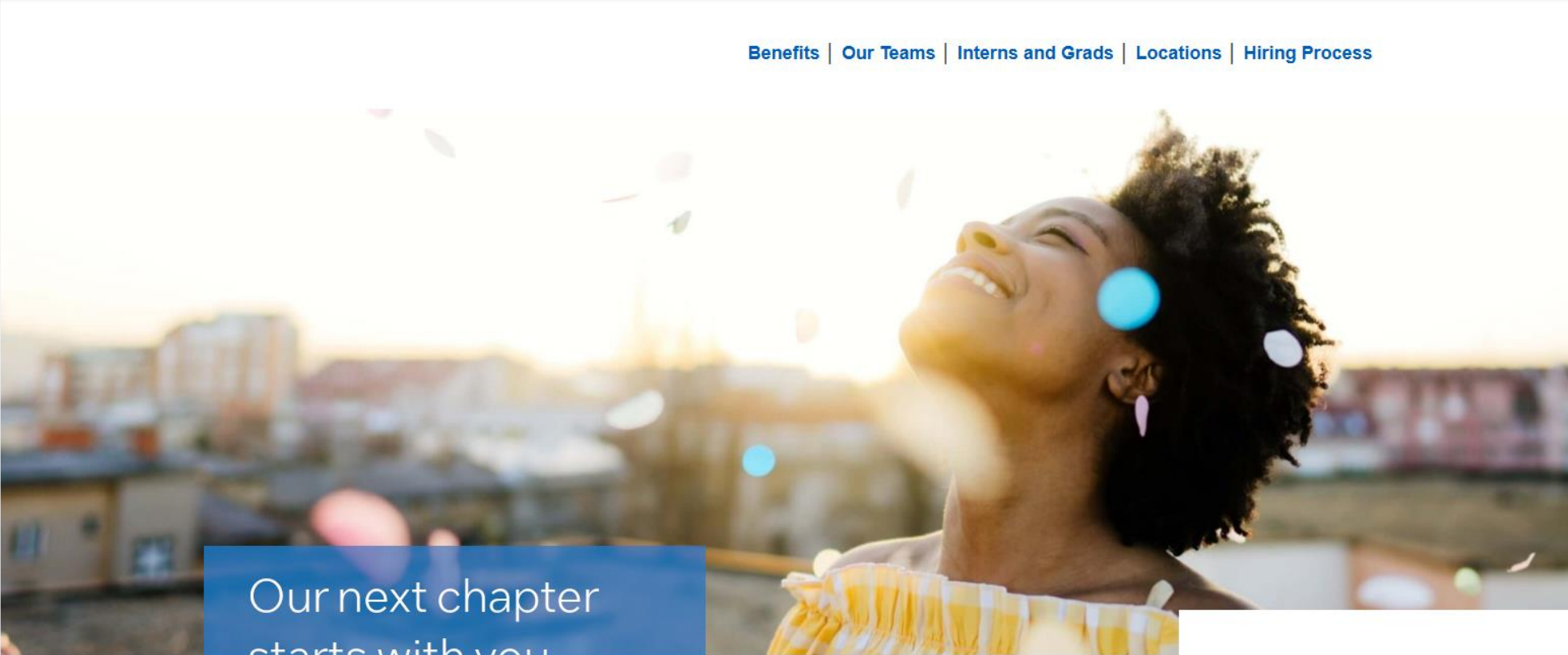
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Job ID: JR0204376

Job Category: Engineering Support

Primary Location: Leixlip, IE

Other Locations:

Job Type: Experienced Hire

Deep Learning Hardware Design Engineer

Job Description

The Movidius VPU IP team is looking for a Digital Design Hardware Engineer to develop leading edge logic design's for Movidius VPUs Artificial Intelligence (AI) accelerator products.

In this role you will join the Movidius Hardware team working closely with multiple internal SoC partners, Hardware Validation Engineers and AI architects to define and drive the development of key AI Hardware Intellectual Property.

Your responsibilities will include:

- Responsible for the logic implementation of complex design block(s) using RTL coding techniques
- Responsible for RTL quality signoff activities such as LINT and CDC checking
- Working with pre-Silicon validation engineers to develop cluster level directed/random tests and environments
- Working with the Physical Design (Layout) team on Synthesis, Formal Verification and Timing Convergence
- Working with FPGA and Emulation teams on early design bring-up to enable SW development.

Qualifications

Qualifications:

- Educational Requirement: Hons. Bachelors/Masters in Electronic Engineering /Computer Science or equivalent

Preferred Requirements:

- Ability to work well in a diverse team environment.
- Exposure to RTL level Digital Design.
- A background in RTL level Digital IC Design using System Verilog and/or Verilog.
- ASIC and/or SoC design experience.
- Experience in unit level validation environment development, test plan generation and test case implementation for the verification of design block(s).
- Experience with industry standard development tools and methodologies.
- Experience with languages such as SystemC, C++, Perl, TCL, Shell scripting, Python, System-Verilog?
- Experience with SoC based CPUs, NoCs, AMBA Protocols, Memory controllers.
- Experience with online collaboration tools
- 1 to 5 years of relevant industry.

Leixlip is the primary location for this position, however, we offer flexibility to work from other Intel sites in Shannon and Cork.

In addition, we are happy to support flexible work/life balance arrangements.

Inside this Business Group

The Internet of Things Group (IOTG) consists of ~4700 dedicated employees across 25 locations across the world. IOTG drives significant growth in the Internet of Things (IoT) market, capitalizing on the inflection and creation of edge computing, leveraging smart, connected devices and artificial intelligence (AI) to enable partners to transform their businesses. IOTG's vision is to lead the industry and enterprises to enhance the way we live and work by making it easier to create exciting, new IoT/edge solutions. IOTG's strategy includes creating unique differentiated silicon to consolidate workloads at the Edge, investing in video and AI inference as the killer app for the Edge, creating new developer tools and ecosystems, and establishing a dedicated vertical consultative sales team to add value to our partners across many verticals.

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We engineer innovative products, while caring for our people and empowering them to reach ambitious goals. We offer work-life balance, long term growth and a supportive environment where individuality is celebrated.



Career Service

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Career Service e orientamento al lavoro

Il Career Service è la struttura dell'Ateneo che aiuta chi ancora studia e chi ha da poco conseguito il titolo a orientarsi nel proprio percorso professionale.

Attraverso laboratori, consulenze, eventi con le aziende e portali per la ricerca di offerte di lavoro e opportunità, fornisce supporto per muovere i primi passi nel mondo del lavoro.



Career Labs

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I Career Labs sono laboratori strutturati in maniera interattiva ed esperienziale. Partendo dai principi teorici, i partecipanti sono stimolati a sperimentare direttamente quanto appreso con attività in aula, che si avvalgono di materiali multimediali come video, modelli ed esercizi su cui lavorare singolarmente e in gruppo.

I Career Labs sono suddivisi in gruppi tematici. Anche se ogni Career Lab può essere seguito singolarmente, consigliamo di seguire tutti i laboratori per avere una visione completa dei temi trattati.

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Our mission is to advance computer architecture and computing systems research and development across Europe. To achieve this, HiPEAC strives to:

- **Strengthen European computing:** advance computing systems as a discipline to reinforce Europe's strategic position in technologies critical for modern society.
- **Nurture talent:** prepare the next generation of world-class computing systems scientists and engineers through academic and professional development.
- **Build a dynamic ecosystem:** unite European research, industry, SMEs, and policy to create a vibrant environment for computing systems design and implementation.
- **Align research & impact:** identify long-term challenges in computing systems, articulate their societal impact, and guide research efforts for greater effectiveness.

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Inria Lyon, France

Post-Doctoral Researcher F/M Dynamic Parallelization of Sparse Codes for High-Performance Computing and Machine Learning



Lyon • [object Object] • Deadline: Oct 31, 2026

Norwegian University of Science and Technology (NTNU), Norway

PhD Candidate in Efficient Edge Intelligence Models



Trondheim • [object Object] • Deadline: Sep 23, 2026

Paderborn University, Germany

PhD position in System Software for Memory-Centric Computing



Paderborn • [object Object] • Deadline: Sep 30, 2026

Paderborn University, Germany

PhD position in Dependable Operating Systems and Runtime Systems



Paderborn • [object Object] • Deadline: Sep 30, 2026

Free University Brussels (VUB), Belgium

Postdoctoral Researcher in Secure and Efficient System Software



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Student Internships

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ESA offers student internship opportunities from three to six months to students in their final or second-to-last year of a Master's degree in technical and non-technical domains. Learn everything you need to know in this article.



European Space Agency



Student Opportunities

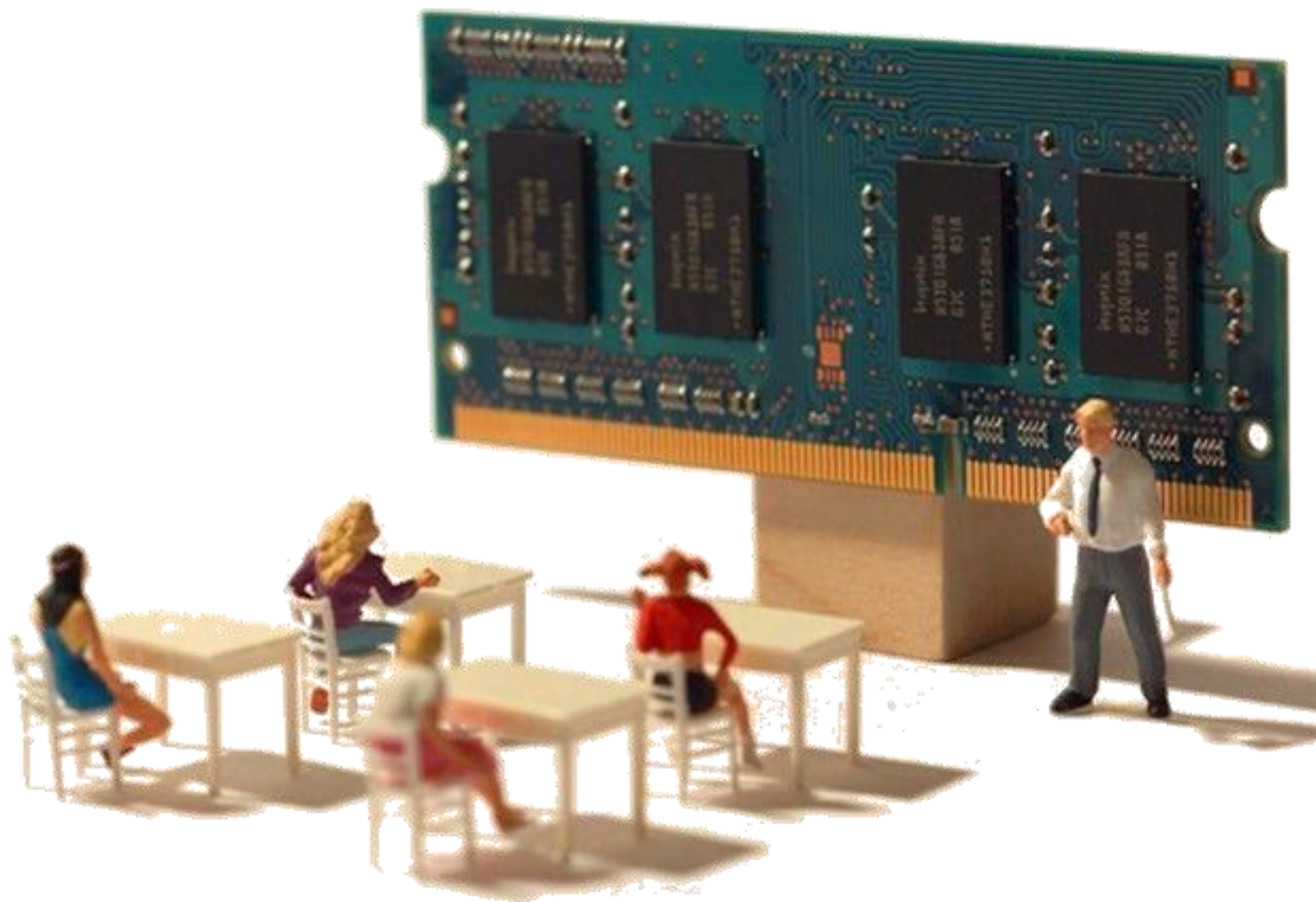
 Search job openings, e.g. "manager"

TECHNICAL STUDENT PROGRAMME

This programme is now open for applications until 3rd March 2025.

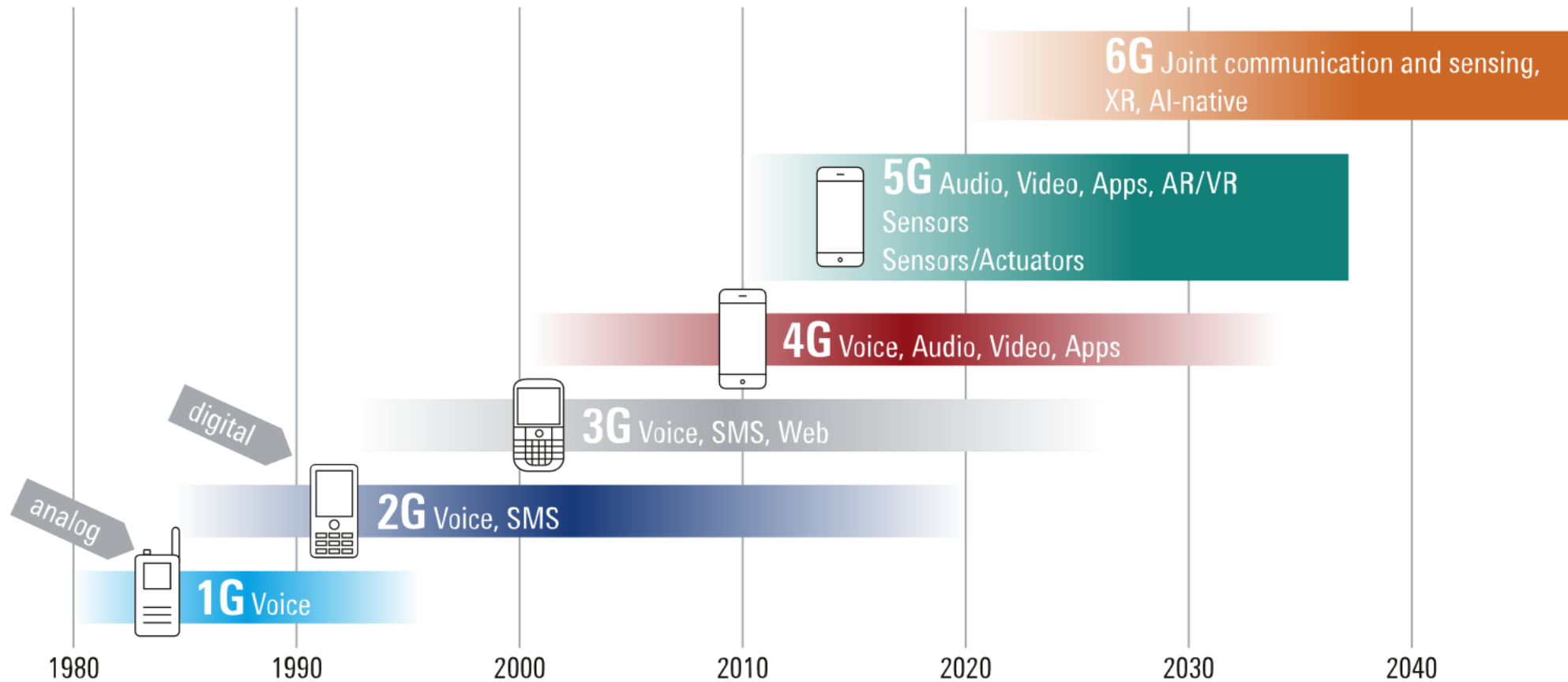
There's no better way to learn than on-the-job. When that job happens to be in a world-famous organisation and center of scientific excellence, even better. If you're an undergraduate in Applied Physics, Engineering or Computing and are looking for a practical training period or a place to complete your final project, you could spend 4 to 12 months at CERN during the course of your studies (Bachelor or Master). An extension of up to a maximum of 14 months may be given.

Questions ?



Let's continue !!

A brief history of mobile communication: from 1G to 6G



Telecom Generation

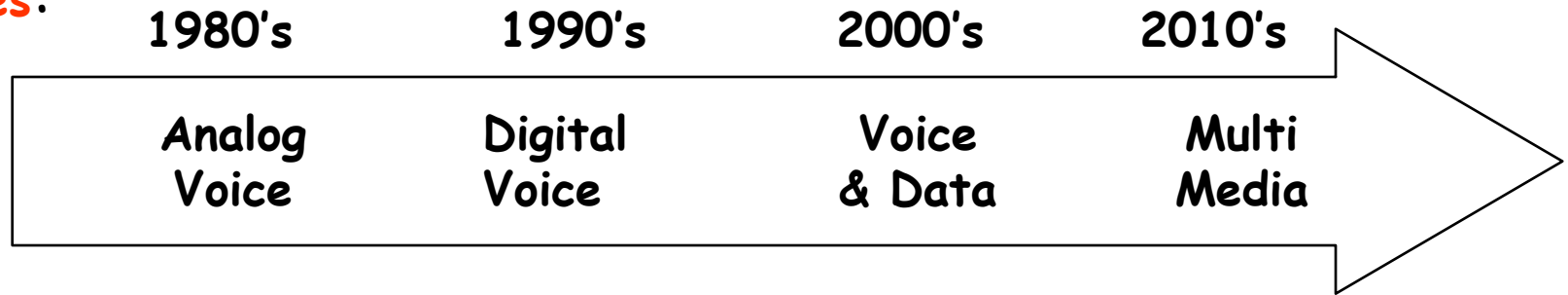
2G 2.5G 3G 4G 4.5G 5G 6G

By Abhijeet Kumar

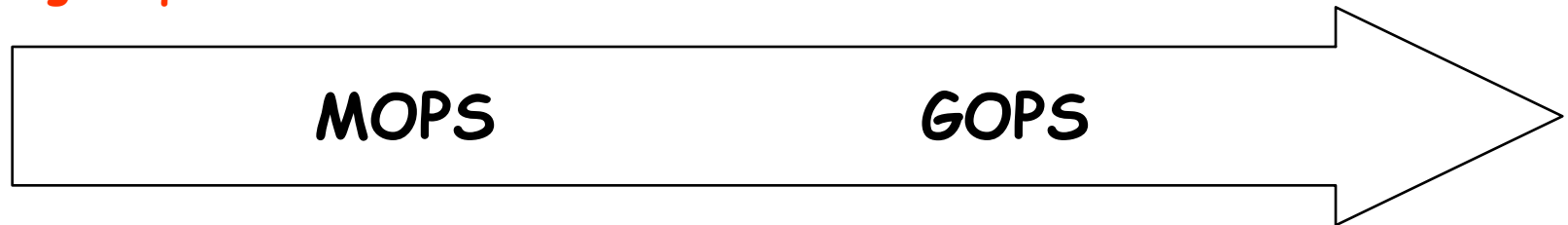
Generation	Technology	Spectrum	Year of Launch	Peak Throughput	Download Speed	Use Cases	Network Architecture	Core Architecture	Latency	Hardware Design	Spectrum Efficiency
2G	GSM	900/1800 MHz	1991	9.6 Kbps	9.6 Kbps	Voice Calls, SMS	Circuit-switched	Circuit-switched	300-1000 ms	Analog/Digital Hybrid	Low
2.5G	GPRS	900/1800 MHz	Late 1990s	40-50 Kbps	35-40 Kbps	MMS, WAP, Email	Packet-switched overlay	Circuit-switched	500-1000 ms	Digital	Slightly improved
3G	UMTS, CDMA2000	1.9 GHz (UMTS), 1.25 MHz (CDMA2000)	2000-2001	2 Mbps	384 Kbps-2 Mbps	Mobile Internet, Video Call	Packet-switched	Packet-switched	100-500 ms	Digital, Smaller Devices	Medium
4G	LTE	700 MHz-2.6 GHz	2008	1 Gbps (LTE-Advanced)	100 Mbps-1 Gbps	High-speed Internet, HD TV, Video Conferencing	IP-based	All-IP (EPC)	10-50 ms	High Speed Packet Access (HSPA)	High
4.5G	LTE Advanced Pro	450 MHz-6 GHz	2014-2015	3 Gbps	1-3 Gbps	IoT, AR/VR, UHD Video	IP-based	All-IP (EPC), Introduction of network slicing	<10 ms	Advanced MIMO, Beamforming	Very High
5G	New Radio (NR)	600 MHz-100 GHz	2017 NSA 2018 SA	20 Gbps	1-20 Gbps	Autonomous Vehicles, Smart Cities, IoT, AR/VR, Remote Surgery	SDN/NFV-based	Service-based architecture	1 ms	Massive MIMO, Beamforming, mmWave	Extremely High
6G	Undefined Yet	Terahertz Band (100 GHz-10 THz)	2028	100 Gbps – 1 Tbps	10 Gbps – 100 Gbps	Advanced AI Services, Extended Reality (XR), Holographic Communications	Undefined Yet	RAN+Core Multivers	<1 ms	Undefined Yet	Undefined Yet

Wireless Market Forces (1/2)

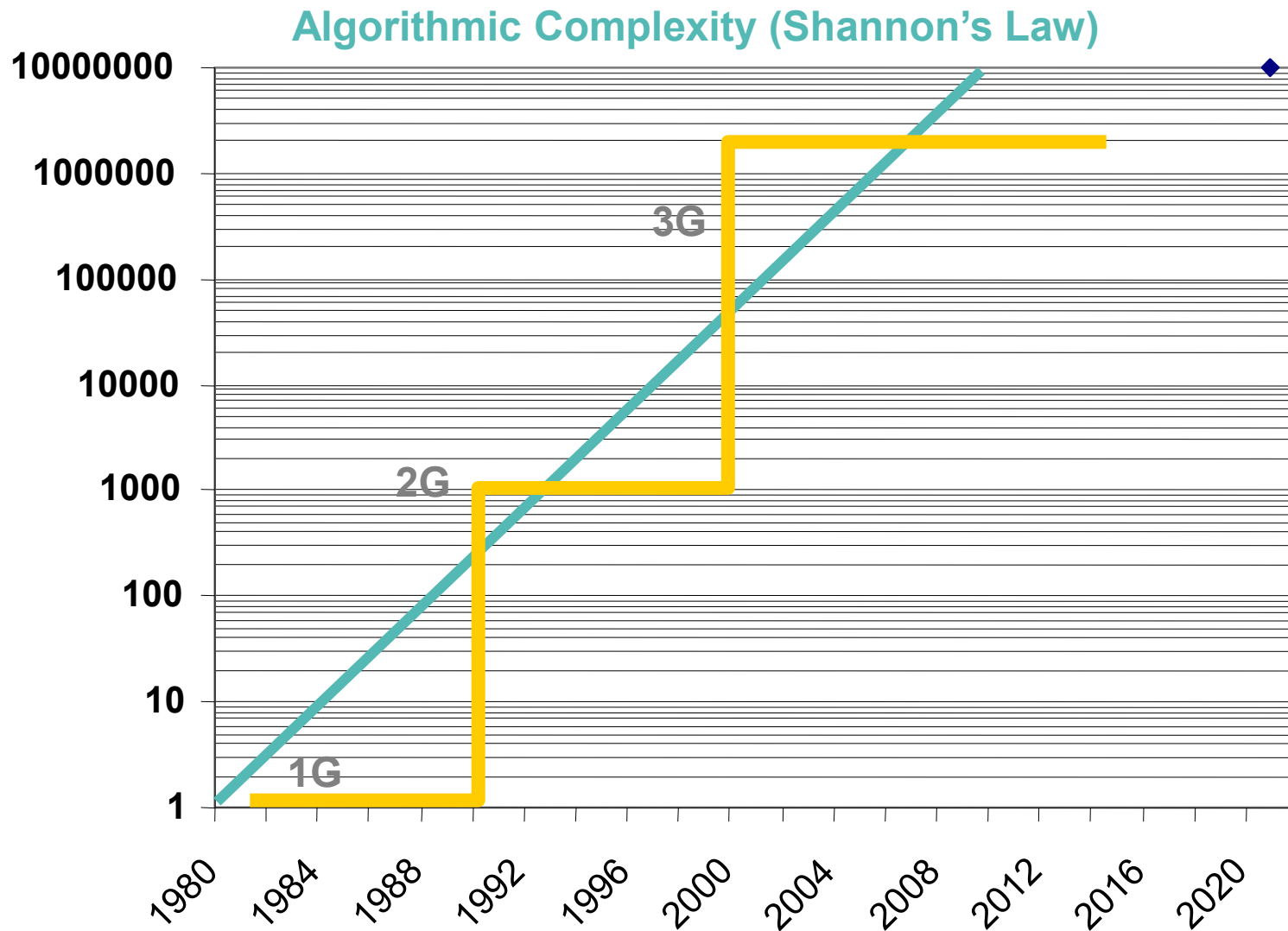
Services:



Processing Requirements:



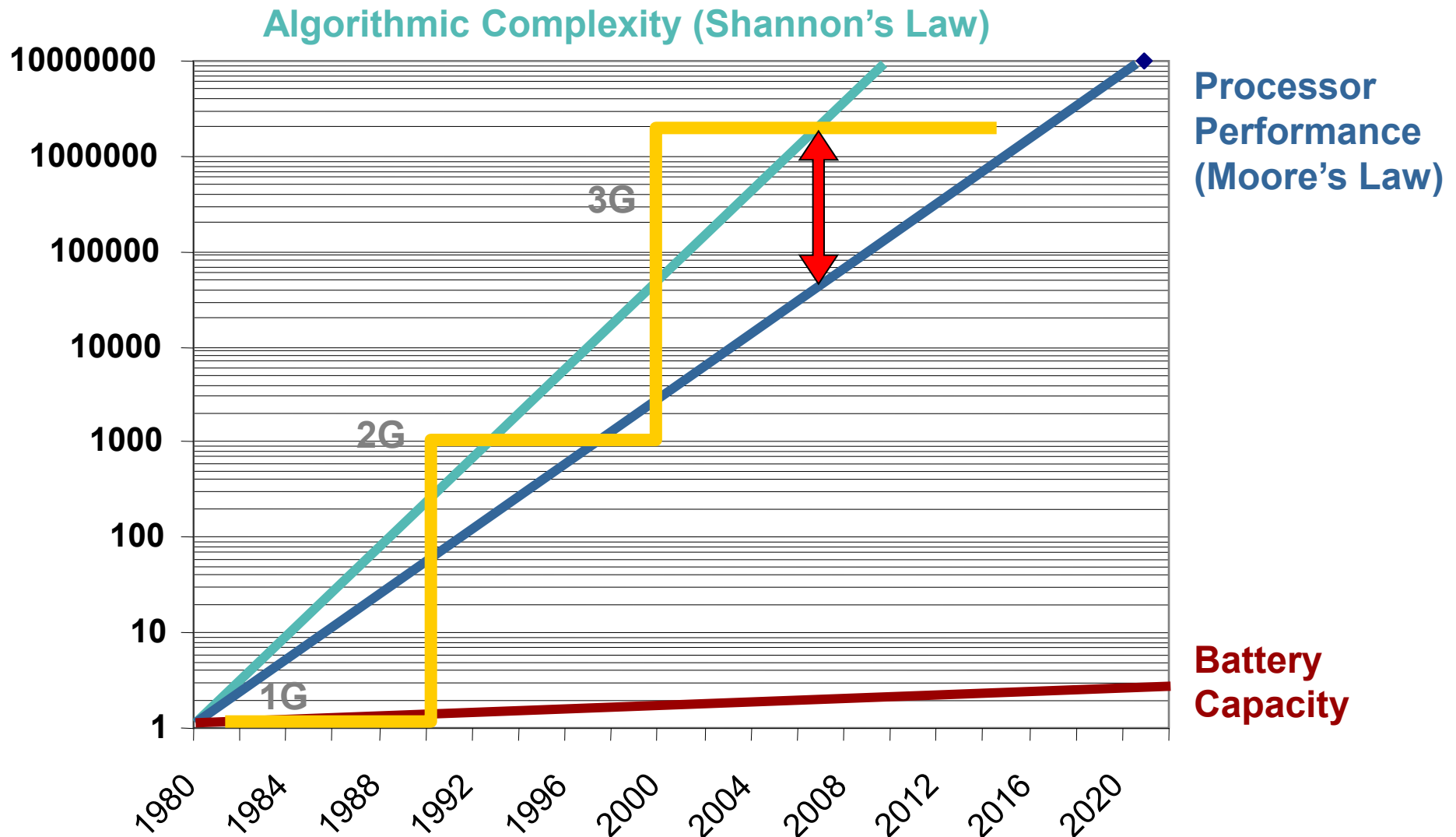
Wireless Market Forces (2/2)



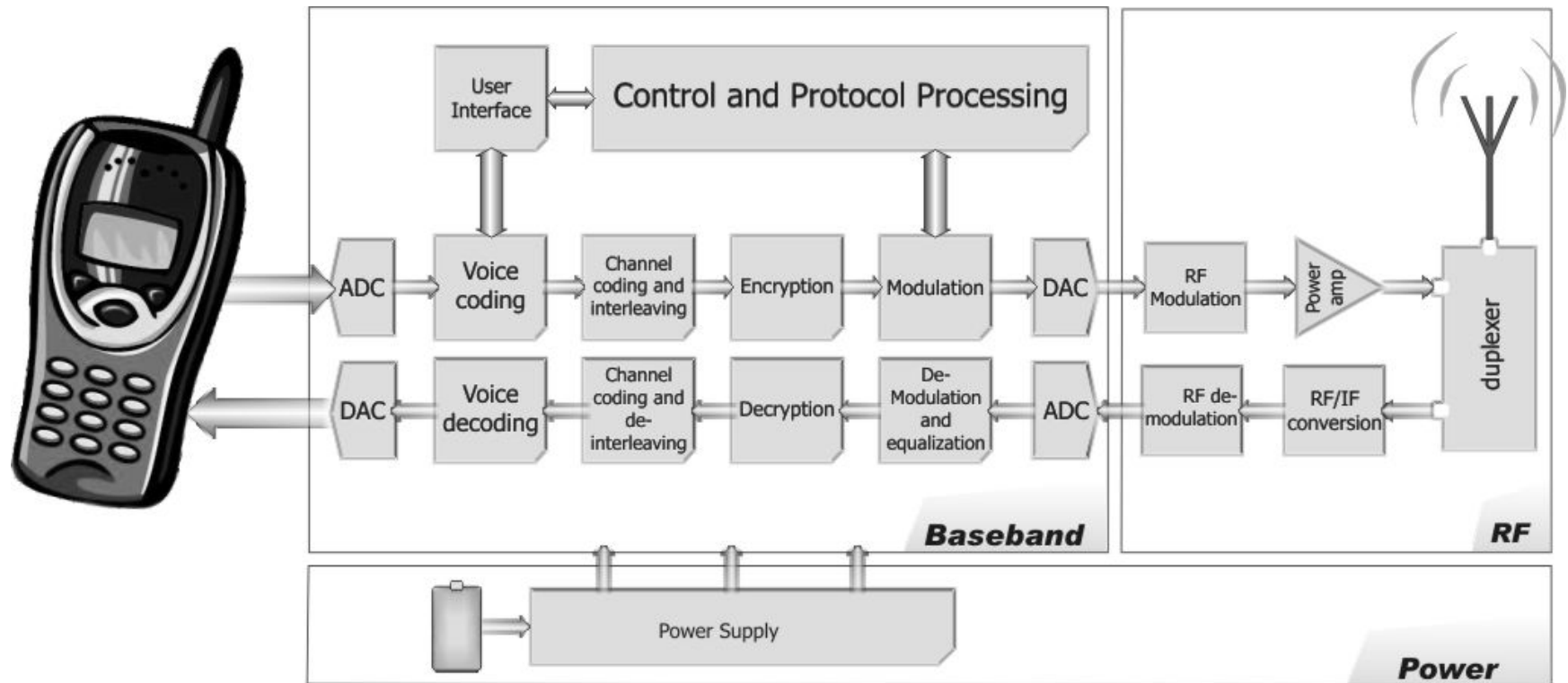
Technology Enablers

- ▣ **More complex communication protocols**
 - ▣ greater need for control processing capability
- ▣ **More sophisticated signal processing:**
 - ▣ interference cancellation
 - ▣ multiple antennas
 - ▣ iterative decoding
- ▣ **Higher levels of sub-system integration**
 - ▣ to bring the cost down
 - ▣ meet performance targets

Wireless Market Forces (2/2)

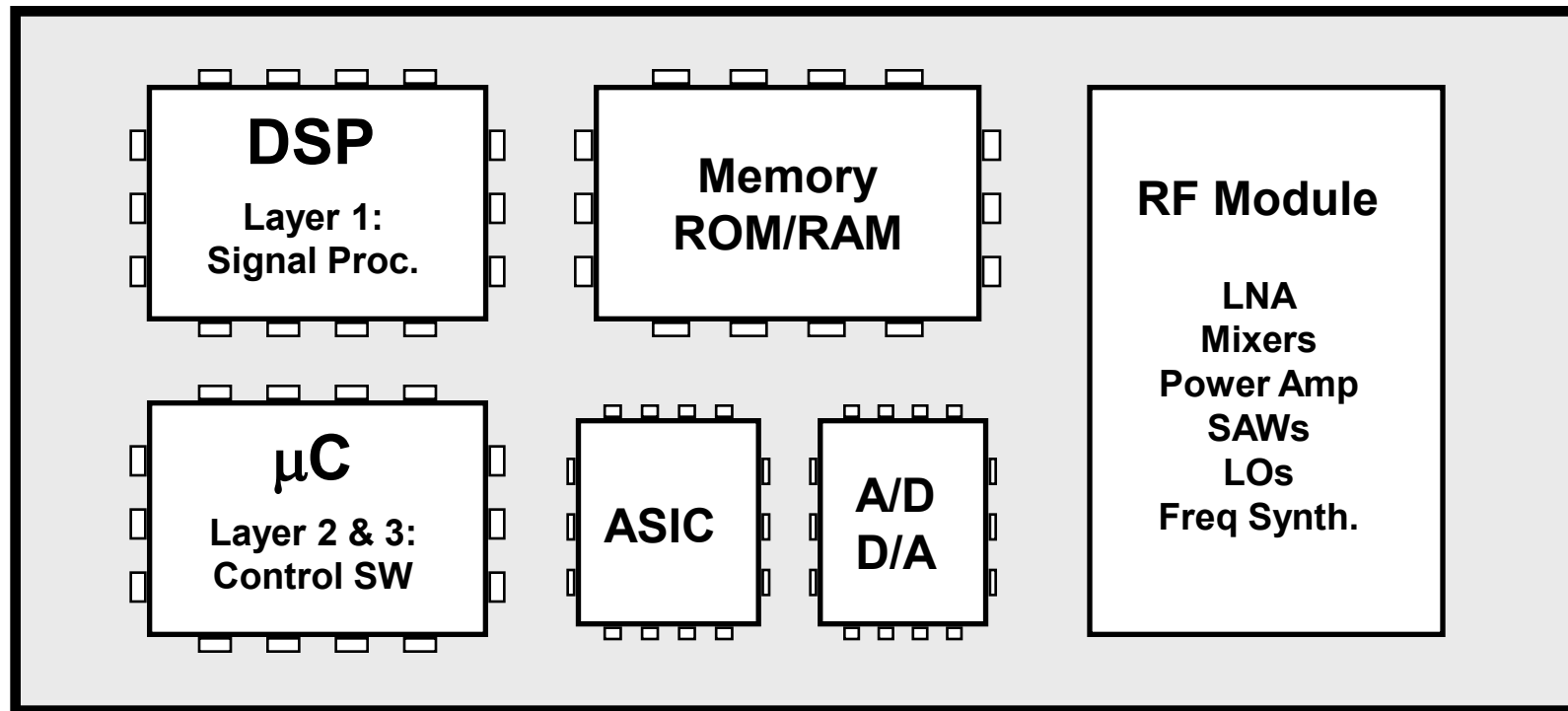


GSM Receiver Implementation (1/3)

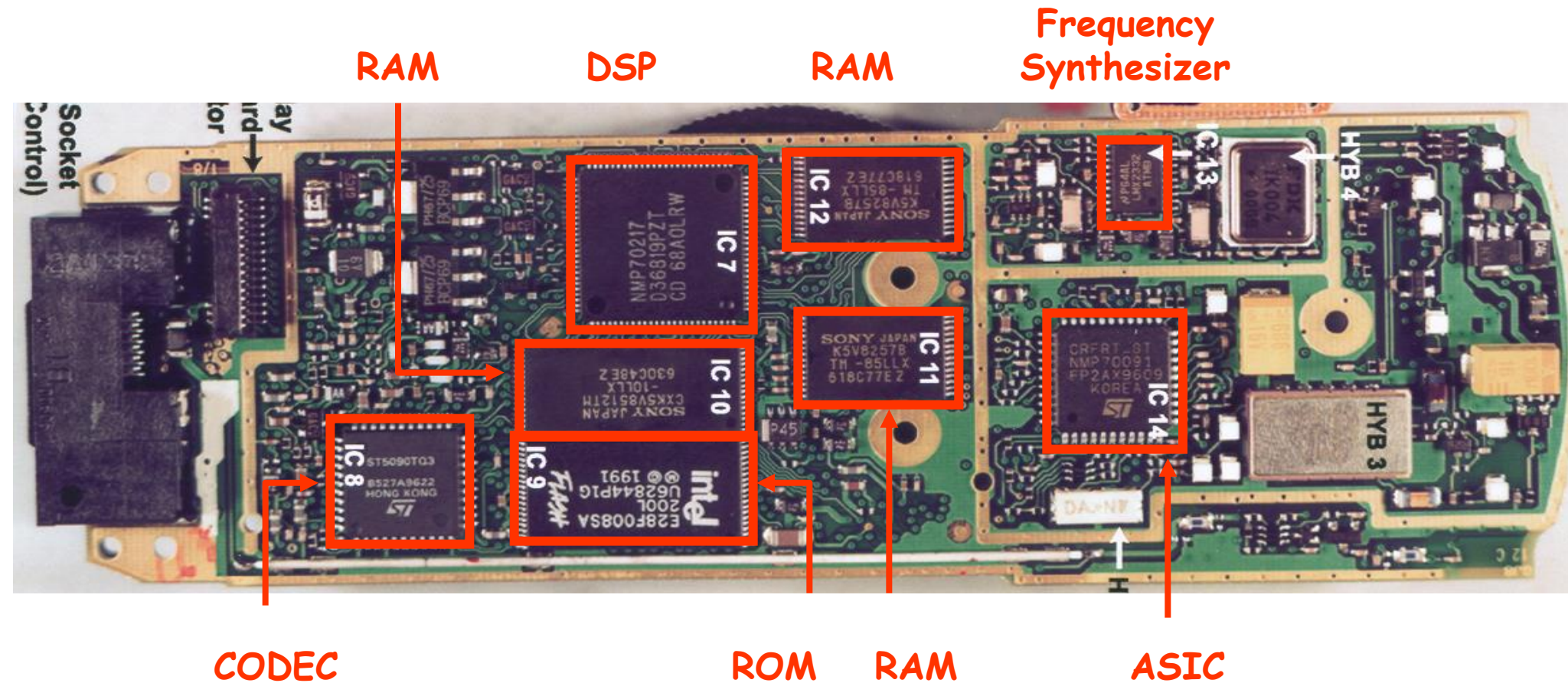


GSM Receiver Implementation (2/3)

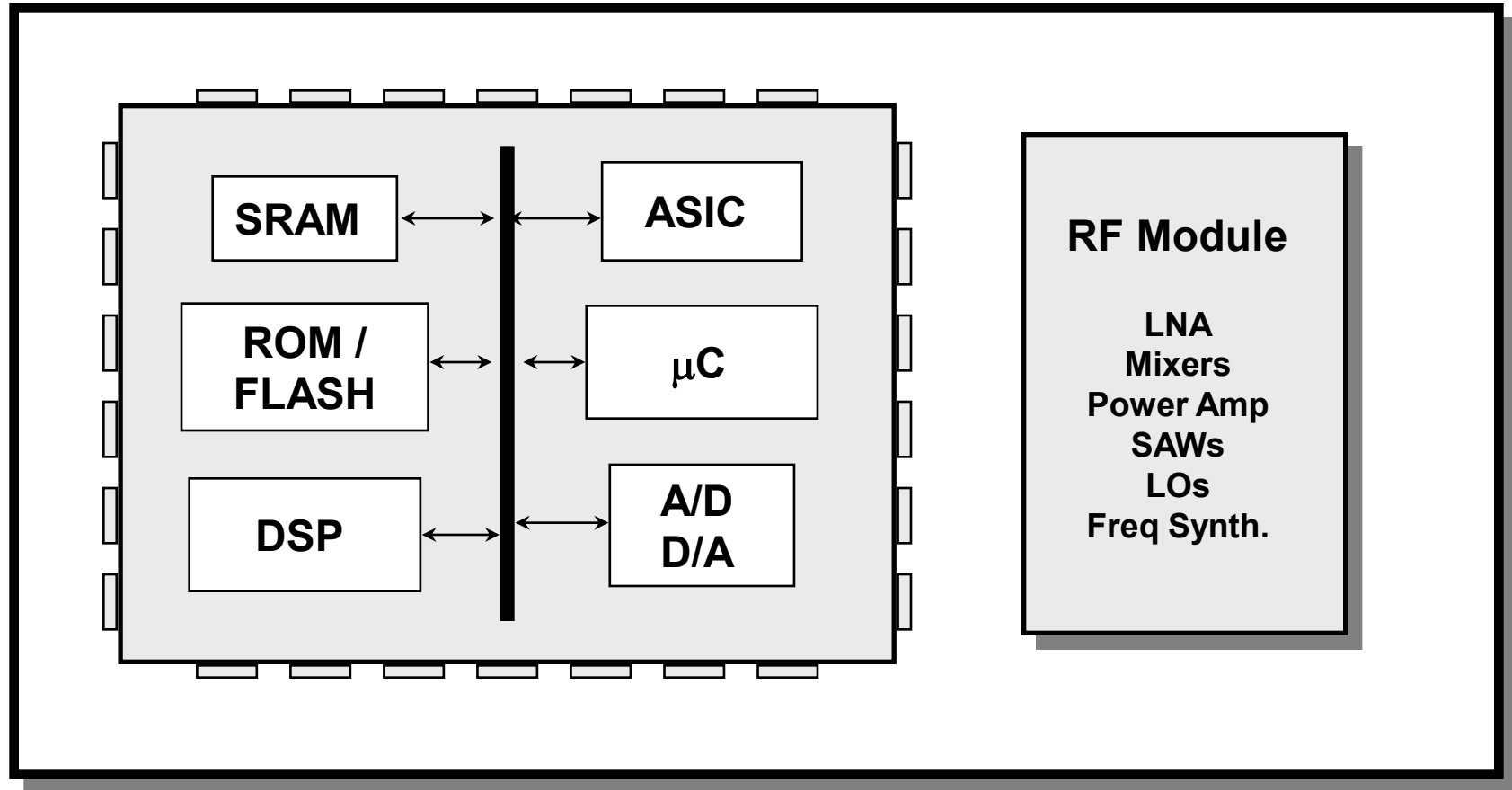
DSP - centric Solution



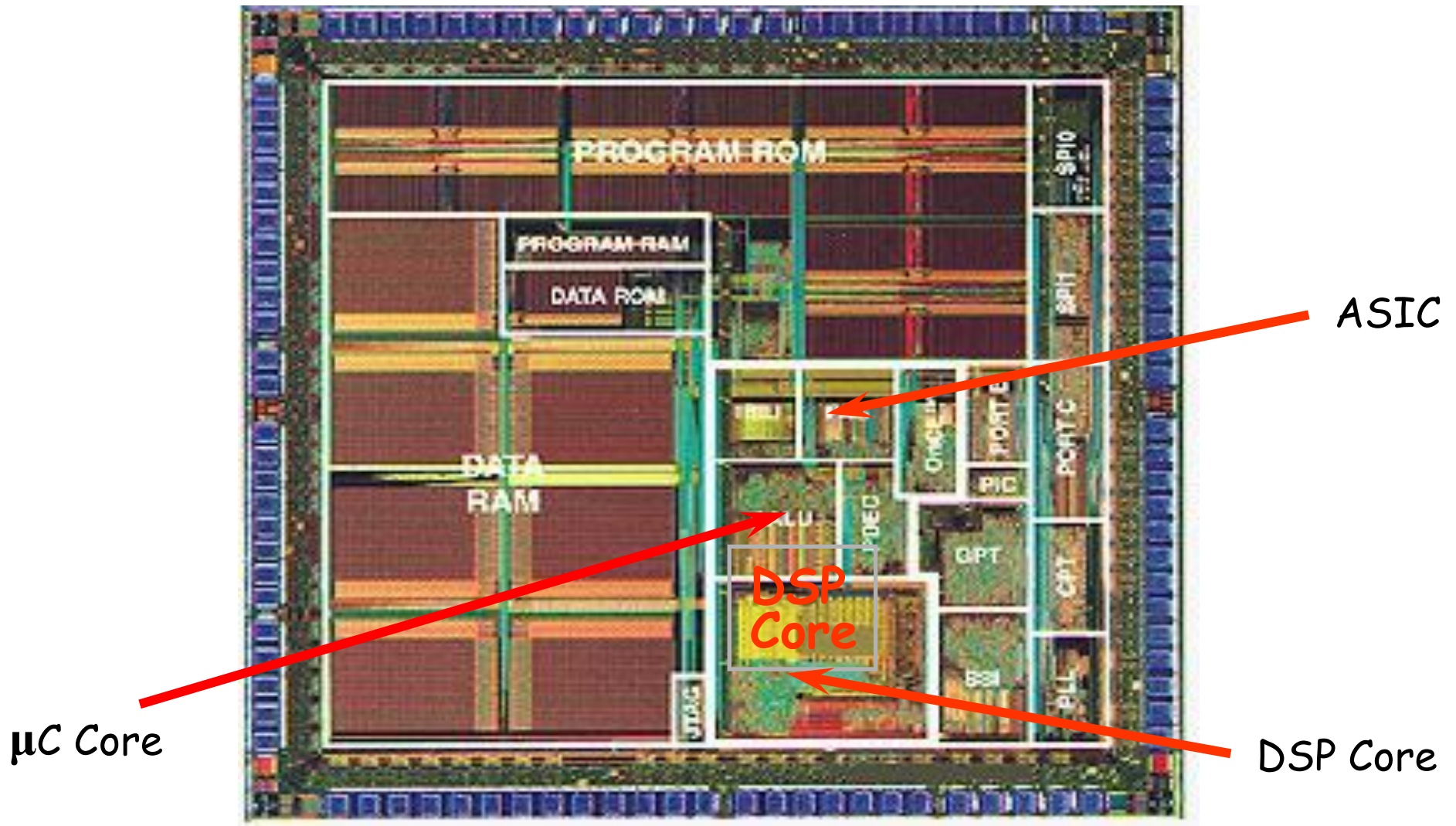
GSM Receiver Implementation (3/3)



GSM SoC Receiver (1/2)

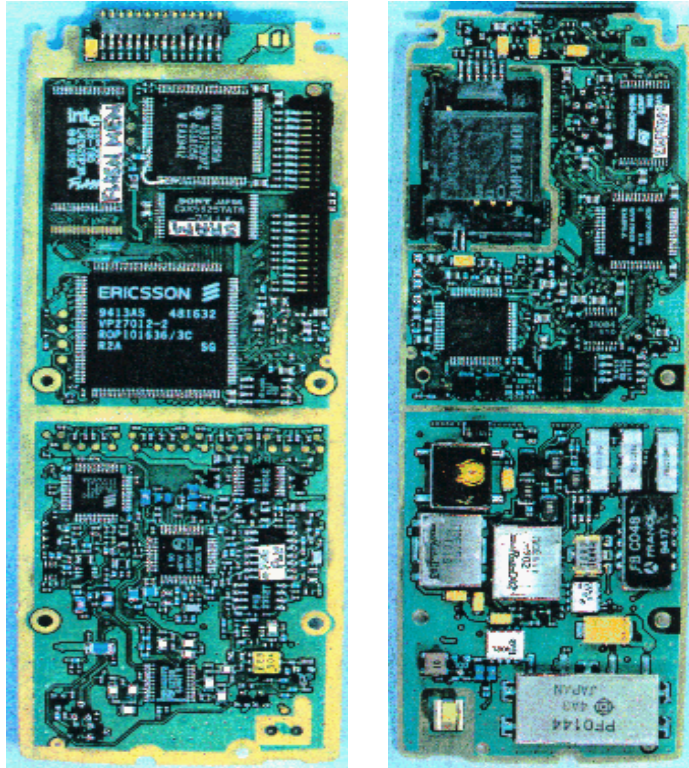


GSM SoC Receiver (2/2)

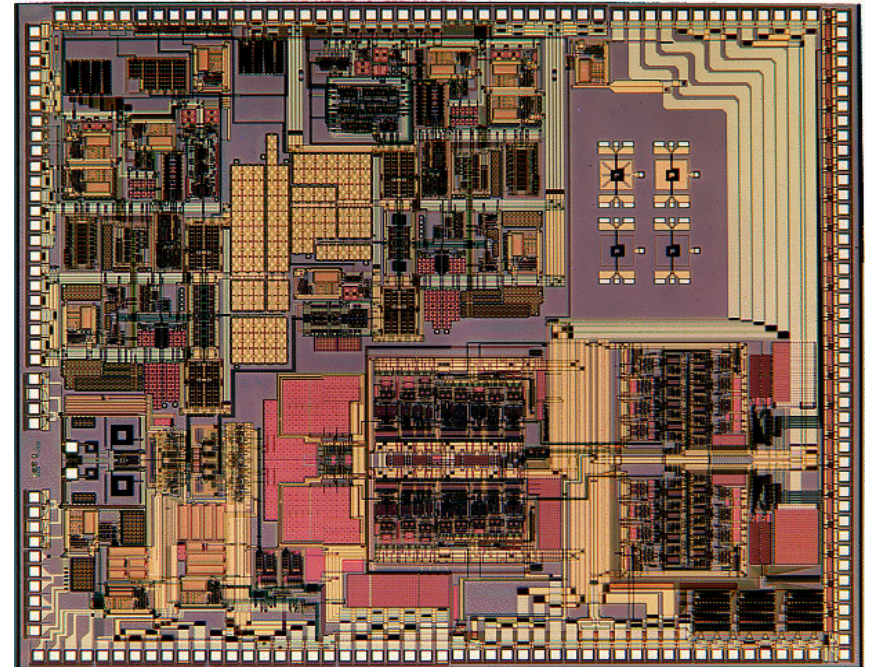


System on a Chip: GSM System

Source:
BWRC

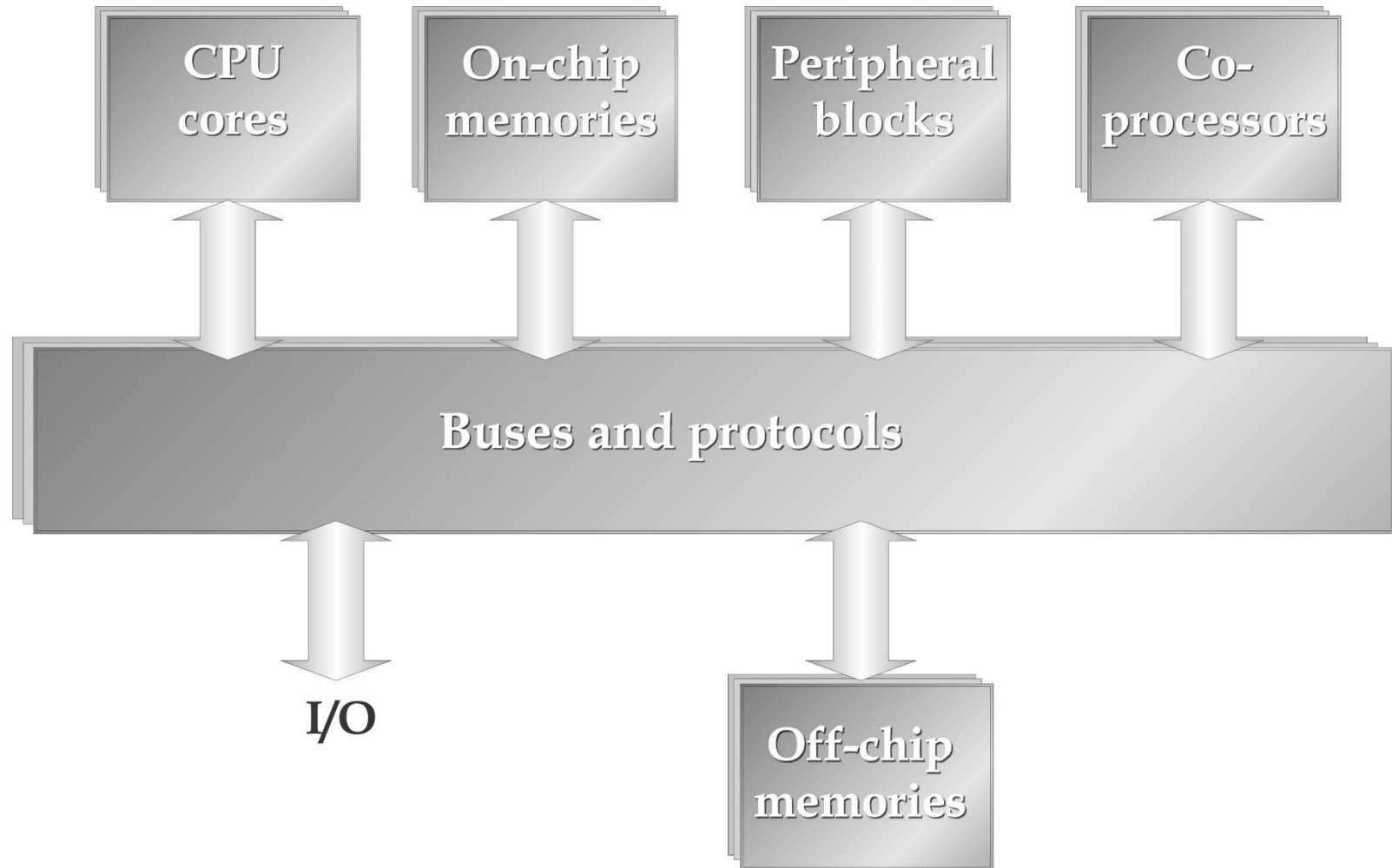


Conventional cellular
phone solution

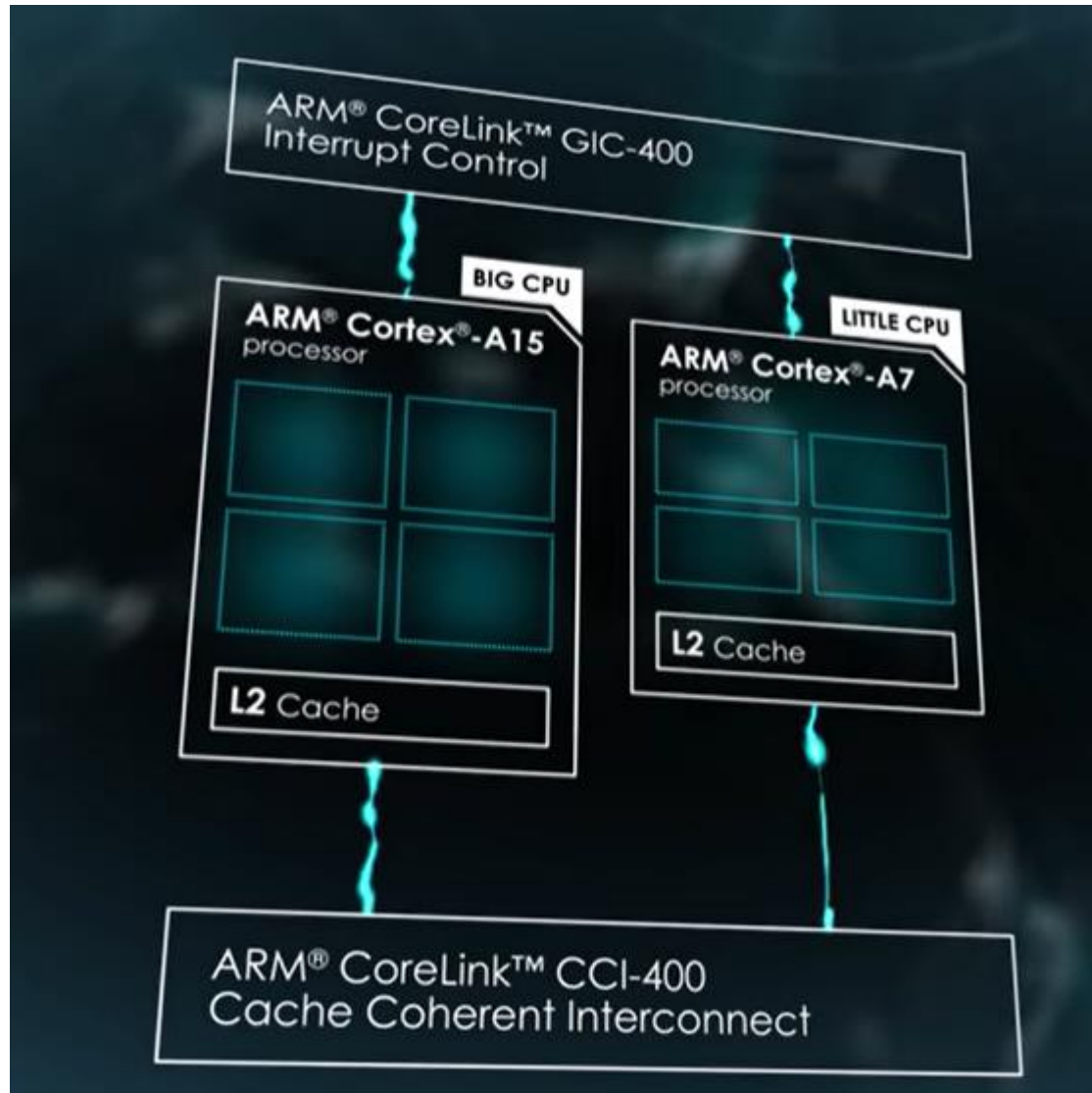


Research into technology and
design methodologies for
CMOS single chip radios

Typical heterogeneous System-on-Chip platform



Example: ARM Big.Little Technology



Key issues (1/3)

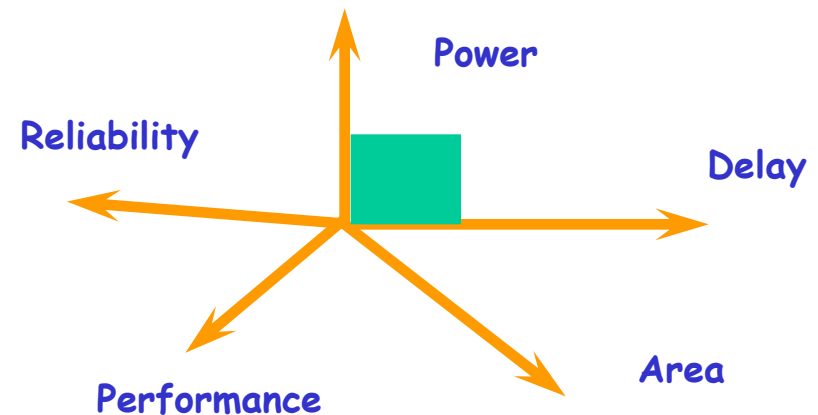
- **how to map** the system/algorithm onto the various building blocks of a heterogeneous, configurable SoC architecture (hardware and software) within given constraints of cost and time to market.
- An extensive **profiling/analysis of the application/algorithm** in the early algorithmic design phases can help to determine the required bounds on performance and flexibility, or to outline the dominant computational pattern and explore data transfer and storage communications.
- This step is both tedious and error prone if carried out **fully manually**, and so **new design methodologies** have to be provided to bridge the gap between **algorithmic development and cost effective realization**.

Key issues (2/3)

- There is a need for fast guidance and early feedback at the algorithm level, without going all the way down to assembly code or hardware layout (thus getting rid of long design cycles).
- Only when the design space has been sufficiently explored at a high level, and when a **limited number of promising candidates have been identified**, a more thorough and accurate evaluation is required for the **final hardware/software partitioning**.
- Most importantly, the optimum system is always the result of a joint, truly, **interactive architecture-algorithm design**.

Key issues (3/3)

- ▣ A better algorithm (even the best) from a communication performance standpoint may not correspond to a suitable computational/communication architecture.
- ▣ Since no single designer can adequately handle algorithms, design methodologies and architectures, a **close interaction** between designers (the system/communication engineer and the VLSI/chip architect) and design teams is required **to master such a complex SoC design space**.



Questions ?

